

Individual Assignment (30%)

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Learning Outcomes Assessed					
CLO3	Demonstrate capacity to perform a self-directed piece of practical work that requires the application of Natural Language Processing and text mining techniques. (P5, PLO6)				

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1.0 Business Concept Development

This section presents the business concept of StudyHub Cafe, a new, data-driven study cafe, where students and professionals will have an opportunity to concentrate on studying, working collaboratively, and achieve success. It encompasses the description of the business, the goals, the vision, the mission, the strategy, and the measurable goals.

1.1 Business Idea

Its business concept is an innovative Study-Centered cafe named StudyHub Cafe. It offers a new kind of environment that allows learners and professionals to reserve private rooms to study, work, and/or collaborate with optional food and drinks. The idea would be to blend the comfort of a cafe and the convenience of a coworking location together with technology and data analytics to provide better customer experience and business performance.

1.2 Business Description

StudyHub Cafe will provide adaptable, hourly based study rooms that will address the requirements of individual and group learners. Customers are afforded the chance to book individual desks, group tables or rooms in advance, depending on their choice and cost. The hourly charges will include RM 4 per hour on individual or group(per person) seats and RM 15 per hour/RM 15 per hour on a private room.

The seating areas have free high speed Wi-Fi, powered outlets, comfortable seats and quiet ambience to bring out the best in one. They can also have access to optional food and beverages like coffee, tea, or light snacks that can be ordered by the customers to supplement their studying time.

All the reservations, transactions, and feedback are kept in digital format and form data that can be analysed by the business to understand the trends, the most active hours, the behaviour of the customers and how long they are staying. The major target consumers would be university students, freelancers and working professionals who require affordable, easy to access and distortion free studying environments.

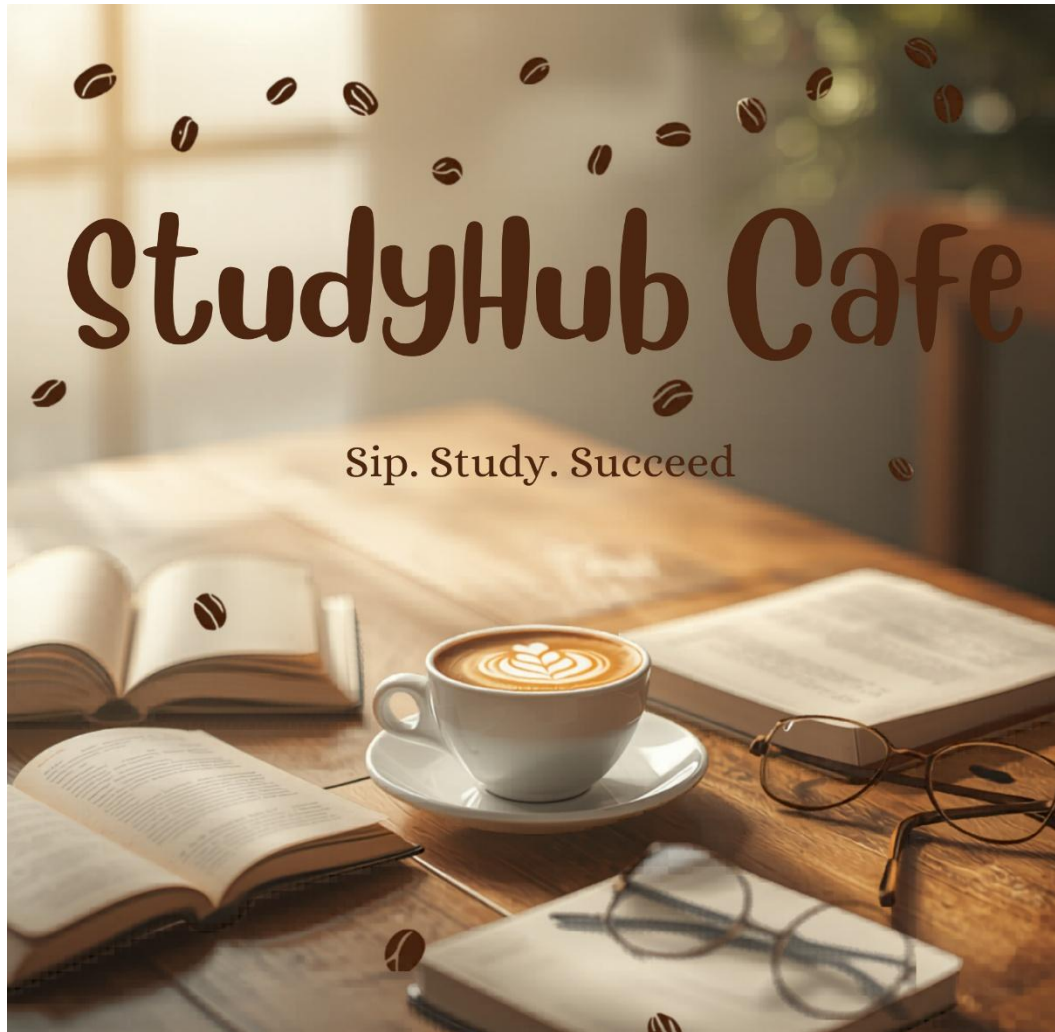
With both technology and comfort, StudyHub Cafe will be developed into a smart environment that facilitates focus and learning. The motto advocated by the concept is **Sip. Study. Succeed**, to position the business as a helpful and fact based study centre as opposed to a commercial cafe.

The StudyHub Cafe provides walk-in and pre booked study. That is because the customers can book their seats online using a reservation system and make sure they are available. Moreover, walk-in customers can also sit down using the seats in case they are idle. This is an adaptable booking platform that enhances customer comfort and maximize seats. The cafe will receive booking information to examine the customer demand, the busiest hours, and the rate of seat occupancy to assist in improving business performance and decision-making.

Another approach that Studyhub Cafe will apply is the loyalty reward scheme where regular patrons will receive points that can be paid in form of free beverages or discounted study time

thus making customers become regulars and maintain the customer base. By bookings, transactions, and customer feedback, the cafe will collect and analyse the information to constantly refine the services and respond to the evolving customer needs. Through a friendly atmosphere plus intelligent information-driven management, the StudyHub Cafe will be the most preferred study area by students and professionals in Nilai, Malaysia.

1.3 Logo Design



Canva was used to create the StudyHub Cafe logo, which symbolizes comfortable and productive studying. The logo is composed of the cup of coffee and books which are a representation of relaxation and learning. The design is simple with the use of simple colors and clean layout to appear professional. The tagline "Sip. Study. Succeed." is a reflection of the mission of the cafe to help students and professionals realize their objectives and have a comfortable place to study.

1.4 Mission, Vision, Strategy, and Goals

In this section, the mission, vision, strategy, and the objectives of StudyHub Cafe are presented. These aspects will give the business a proper orientation and it will also ensure success of StudyHub Cafe in the long run. The mission focuses on the purpose of the business and values, whereas the vision exhibits long-term objectives of the cafe. The

strategy gives a description of how the business is going to realize its objectives based on the data-driven decision making, and the goals included measurable goals to measure business performance and business development.

1.4.1 Mission

StudyHub Cafe will aim to offer comfortable and productive study conditions to the students, freelancers, and working professionals in Nilai, Malaysia. The cafe will provide low cost and flexible study options by provision of separate desks, group tables, and single room that will have high speed Wi-Fi and other facilities. The StudyHub Cafe is very committed to high-quality customer service and constantly enhances its services according to the customer reviews and data analysis. Using the information on booking details and understanding of customers, the cafe attempts to make the overall study process as effective as possible and helps each customer to reach his or her objectives in an environment free of distraction. The cafe also aims at establishing long-term association with its customers in the forms of loyalty programs, student delivery of services, and tailor-made services in order to serve the increasing community needs of its consumers.

1.4.2 Vision

StudyHub Cafe has a vision that it can be the best study based cafe in Nilai and known as a provider of great productivity zones which promote learning and even teamwork. The cafe also sets its goal on growing into various businesses in Malaysia whereby more students and workers have the opportunity to enjoy convenient and affordable environment to study. By pursuing the steady implementation of smart booking systems, data analytics, and customer-oriented changes, StudyHub Cafe hopes to become a model of modern study spaces in Malaysia. The cafe also projects to establish a good relationship with the universities and colleges within the Nilai to have exclusive study packages and co-educational learning by the students. Finally, StudyHub Cafe will aim at becoming the most reliable and favorite place of study where all the customers could sip, study, and succeed.

1.4.3 Strategy

To be a successful business and enhance customer satisfaction, StudyHub Cafe will maintain a data-driven approach to its business growth. The cafe will gather information based on the bookings, customer visits, and customer reviews to determine the times of the day when there is the highest demand and the preferences of the customers. This data will allow the maximization of seating plans, pricing and staffing. StudyHub Cafe would further have the option of pre-booking seats and walking-in to maximize the number of seats.

The cafe will also advertise its services using the social media, student discounts, and loyalty programs. The company will be concentrated on the creation of comfortable and productive atmosphere to encourage the repeat customers. StudyHub Cafe will also have flexible pricing individual, group, and private room. With the ambition of sustainable growth and success in the long-term, StudyHub Cafe is expected to succeed with the help of data-driven decision-making and focusing on the interests of customers.

1.4.4 Goals

- A minimum of 10,000RM monthly revenue during the first year of operation can be achieved by measuring this in terms of monthly sales reports and booking records.
- Attain a rate of 90% customer satisfaction in the initial half-year, which will be gauged by post-visit electronic feedback forms and app ratings which will be taken at the end of every booking session.
- Achieve a minimum occupancy level of 75% during busy periods (afternoon and evening) in the initial six months that are monitored with the assistance of the online booking system and walk-in logs.

2.0 Market Research and Trends

This section will give the market research of StudyHub Cafe in Nilai, Malaysia. The study is targeting customer demand, market trends and business opportunities. Nilai is an emerging educational place and has multiple universities and colleges and thus it is the ideal place where a study oriented cafe should be established. The growth of learners and freelancers in Nilai puts the city under high demand of tranquil and comfortable study space.

2.1 Market Research

To conduct market research related to StudyHub Cafe, a variety of AI and analytics tools were used to obtain appropriate information on customer behaviour, market trends, and business viability in the market of Nilai, Malaysia. Insights into the preferences of customers and the patterns of behaviors among university students and freelancers in Malaysia were generated with the help of ChatGPT. Google Trends was used to examine the search interest on study cafes and coworking space in Malaysia within the last five years indicating a steady positive trend especially during examination times. The statistical data regarding the cafe and coworking space industry growth in Southeast Asia obtained with the help of Statista confirmed the presence of the growing market of the study-oriented enterprises. The business viability of StudyHub Cafe in Nilai was determined using VenturusAI assessing the size of target market, competition, and revenue potential. The next search query was the one employed with AI tools

“Act as a business analyst. I want to start a study cafe in Nilai, Malaysia, and I would like to study the market and trends to assess the feasibility of this idea. Focus on data volume, customer behavior patterns, and digital trends.”

All the results of these tools were tabulated and evaluated to get a full picture of the market situation, customer needs, and competition of StudyHub Cafe in Nilai.

2.2 Summarize Findings

According to the results of market research, the demand of university students and younger specialists in Malaysia in the dedicated spaces to study has been increasing steadily throughout the last several years. Statista showed that it is believed that the pace of expansion of the cafe and coworking space sector in Southeast Asia averaged 8% annually and this serves as an affirmative indication of a robust and growing market that includes the study-

based companies such as StudyHub Cafe. According to Google trend information, key words such as, study cafes and coworking space in Malaysia have started to be searched more and more, especially in April, May, October and November of every year when exams are being held.

Nilai is an upcoming educational centre with a population of more than 21,000 students at five major institutions such as INTI International University, Nilai University, Universiti Sains Islam Malaysia (USIM), MILA and KPJ University College that makes StudyHub Cafe a well-established level of customer base all through the year. Competitor analysis shows that although Nilai has some regular cafes and a local library, none of them are study-oriented with an opportunity to book the study room in advance, be locked and have adaptable hourly prices, giving the StudyHub Cafe a definite substantial competitive edge in the market. The current market size in Nilai is owned by regular cafes, which have about 40% market share of the study space followed by the next 20 percent by public libraries and 10 percent. coworking space with a small gap of 30 percent market share that can be occupied by StudyHub Cafe. The evidence of customer behavior reveals that university students and freelancers have preferred to spend their study time in the evening hours between 6pm and 9pm and then in the afternoon hours between 2pm and 5pm implying that StudyHub Cafe should make maximum seats and staffing at these high times.

Most target customers who include 55% of university students and 25% freelancers will want individual seats at low hourly rates which implies that studying place should focus more on individual desks and still has group and private room where one can share them with friends. The research on customer spending behavior informs that university students in Malaysia have an average expenditure of RM15 to RM30 on a single study session and this will suit perfectly in the pricing strategy of the StudyHub Cafe and prove that the services cost-effective and appealing indeed. It has been also shown that young Malaysians start to prefer digital payment options like e-wallets, and more than half of average transactions will be conducted via such apps as Touch and Go and GrabPay, which will help StudyHub Cafe to implement the digital payment system. Research also indicates that students are even ready to pay to study in comfortable and well-prepared areas especially during the examination whereby the atmosphere in libraries and the home setting can be too noisy or crowded

Altogether, the market research indicates that StudyHub Cafe has great opportunities to prosper in Nilai because of the significant number of students, increasing need in productive studying facilities, the absence of specific competitors, and positive customer behavioral tendencies that perfectly fit into the business model and pricing strategy of the cafe.

2.3 Charts & Graphs

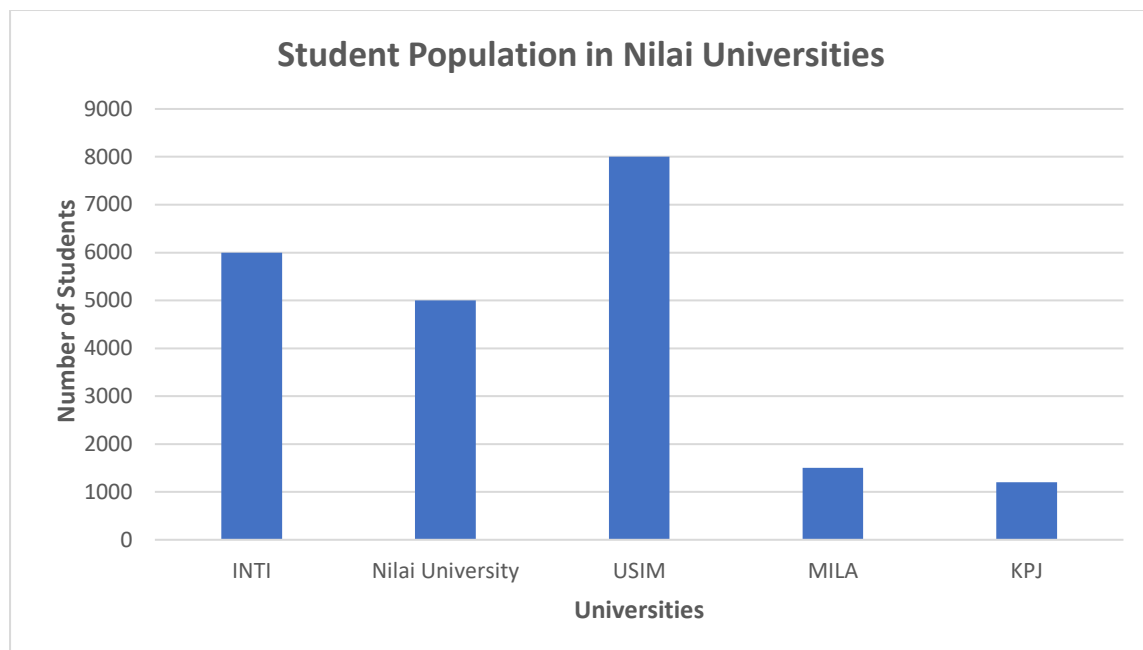


Figure 1: Student Population in Nilai Universities

The bar chart above illustrates the projected number of learners in the big universities in Nilai. The highest student population of about 8,000 students are in University Sains Islam Malaysia (USIM) and 6,000 and 5,000 students in INTI International University and Nilai University respectively. Although MILA and KPJ University College have lower student populations of around 1,500 and 1,200 students respectively, they nevertheless provide contributions to the total demand of study spaces in Nilai. The combination of these five universities being located closely means that there would always be a strong demand of good study locations within Nilai during the academic year. Therefore, the market potential of StudyHub Cafe is high due to the high and varied number of students in the area. A total of more than 21,000 students at the five universities will also give the StudyHub Cafe considerable customer base to maintain its business operations and even expand into the future.

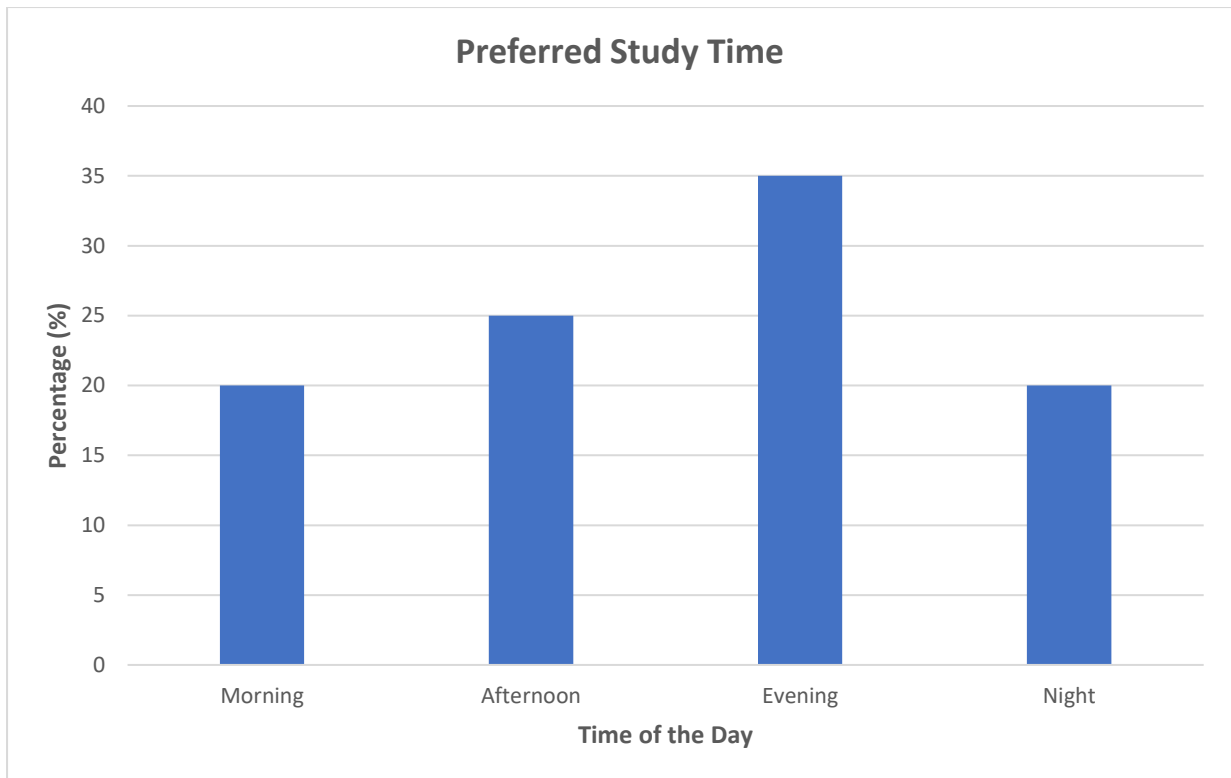


Figure 2: Preferred Study Time of Customers

The chart in the above indicates the time of day the customers prefer to have a study. Evening time is the most preferred at 35% and afternoon at 25%. There is a preference of 20% in the morning and in the night. This means that the StudyHub Cafe will be in high demands in the evening hours. This is why the cafe must reserve additional seats and resources at this period.

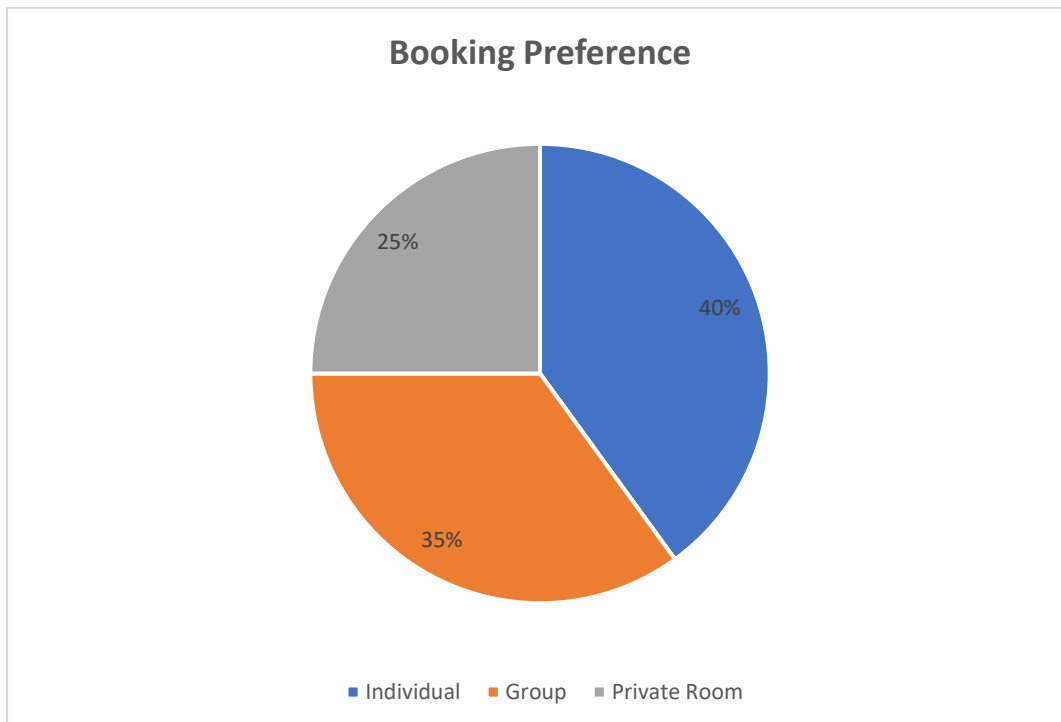


Figure 3: Booking Preference of Customers

In the chart above, preferences of customers in booking at StudyHub Cafe are depicted. The most highly desired seating is individual seating with more than 40 percent and the second least is the group seating 35 percent and lastly the private rooms seating is 25 per cent. This implies that the majority of customers are more inclined to low-cost personal seating. As such, StudyHub Cafe should dedicate additional seating to the customers.

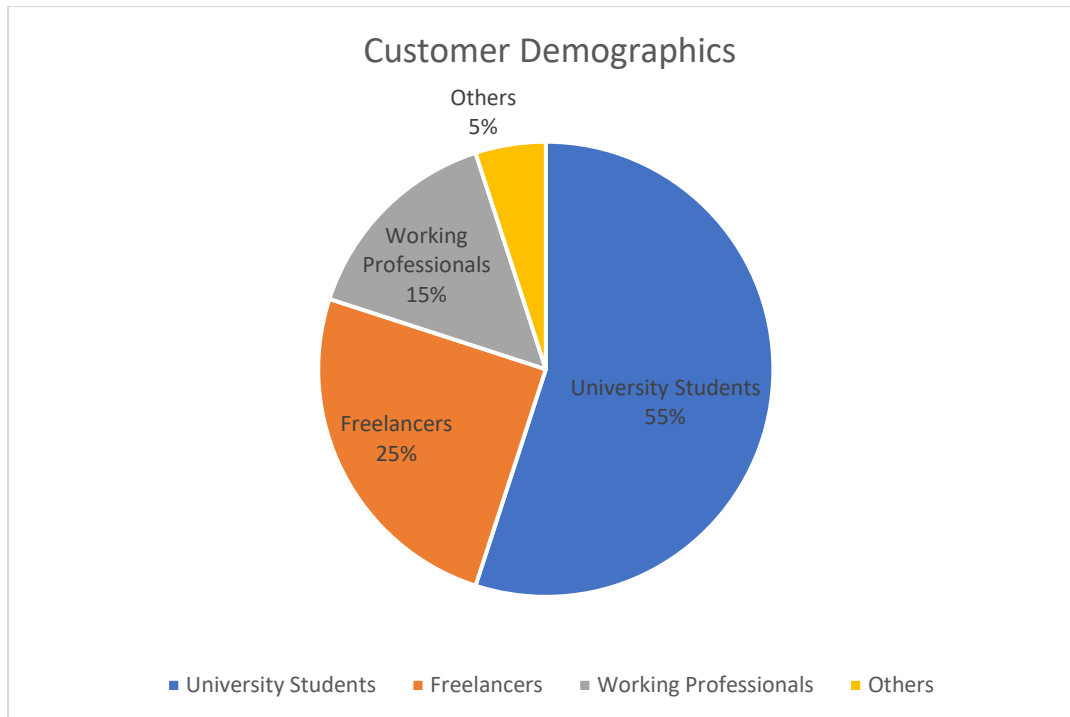


Figure 4: Customer Demographics of StudyHub Cafe

The pie chart above shows the projected demographics of the customers of StudyHub Cafe. Students of the university constitute the biggest group of customers at 55% which implies that the target market of the cafe comprises students of the local institutions like INTI International University, Nilai University and USIM. The second greatest segment is the freelancers who constitute 25 percent due to the increase in remote job and flexible work provisions. Work professionals constitute 15 percent of the customer base indicating that StudyHub Cafe is also attractive to working professionals who need to work in a quiet and productive environment not necessarily at office. The rest 5 percent are the other customers like school students or visitors. These statistics underscore the fact that University students and freelancers should form the core of the marketing campaigns of StudyHub Cafe because they constitute 80 per cent of the overall customer base. The customer demographics will enable the studyhub cafe to customise its facilities, services and prices to serve the needs of the main target customers.

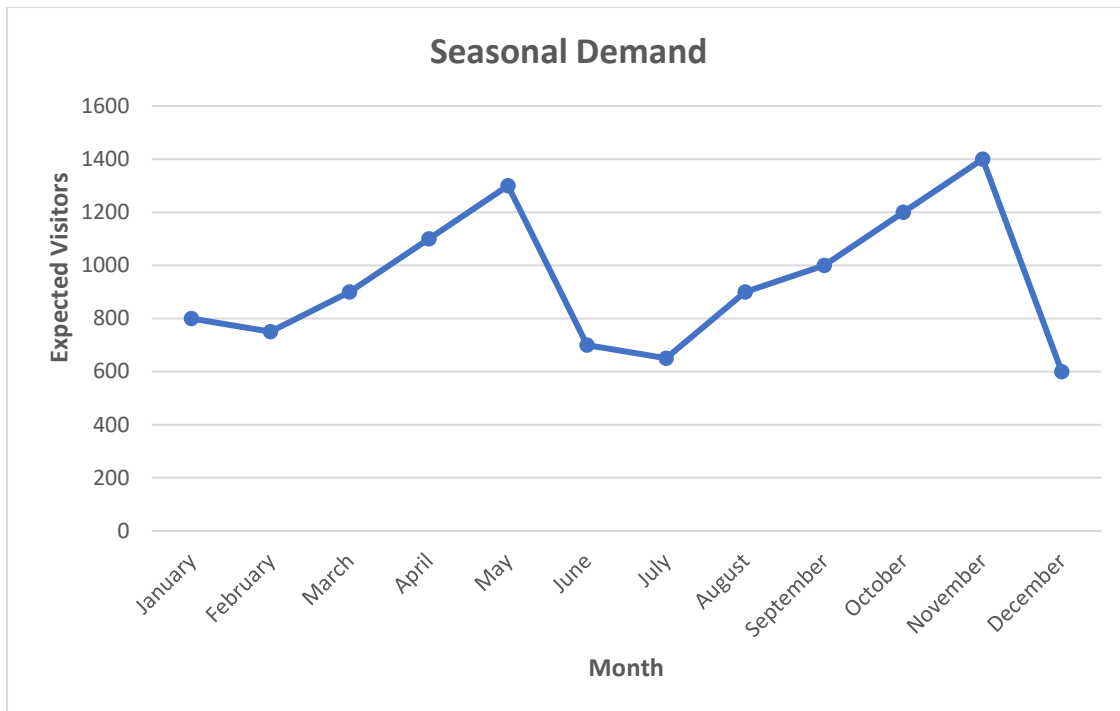


Figure 5: Seasonal Demand of StudyHub Cafe

The above line chart will demonstrate how the monthly visitor trends of StudyHub Cafe are expected to be throughout the year. The information shows that there were two important peaks that appear during exam seasons in the university. The highest one will be experienced in May when there will be about 1,300 visitors and this could be attributed to the fact that most universities are in the end Semester examinations in Nilai. The second and maximum peak is reported during the month of November whereby there are about 1,400 visitors which is the final examination period of the academic year. The visitor numbers decrease drastically in between June and July with about 700 and 650 visitors respectively since these are the semester recessions when students are not present on campus. Equally, the lowest number of visitors is recorded in December, which is around 600 visitors, and that is especially at the end of the year of the holiday. The evidence indicates that studyhub Cafe needs to staff up, lengthen opening hours and make sure that it has as much availability of seats as possible during the peak months like May and November. The cafe may also launch promotional deals and student discounts during off-peak months such as June, July, and December so as to appeal to more customers and ensure that revenue is also stable. Knowing these seasonal trends allows StudyHub Cafe to make evidence-based decisions that would improve operations and the general business performance.

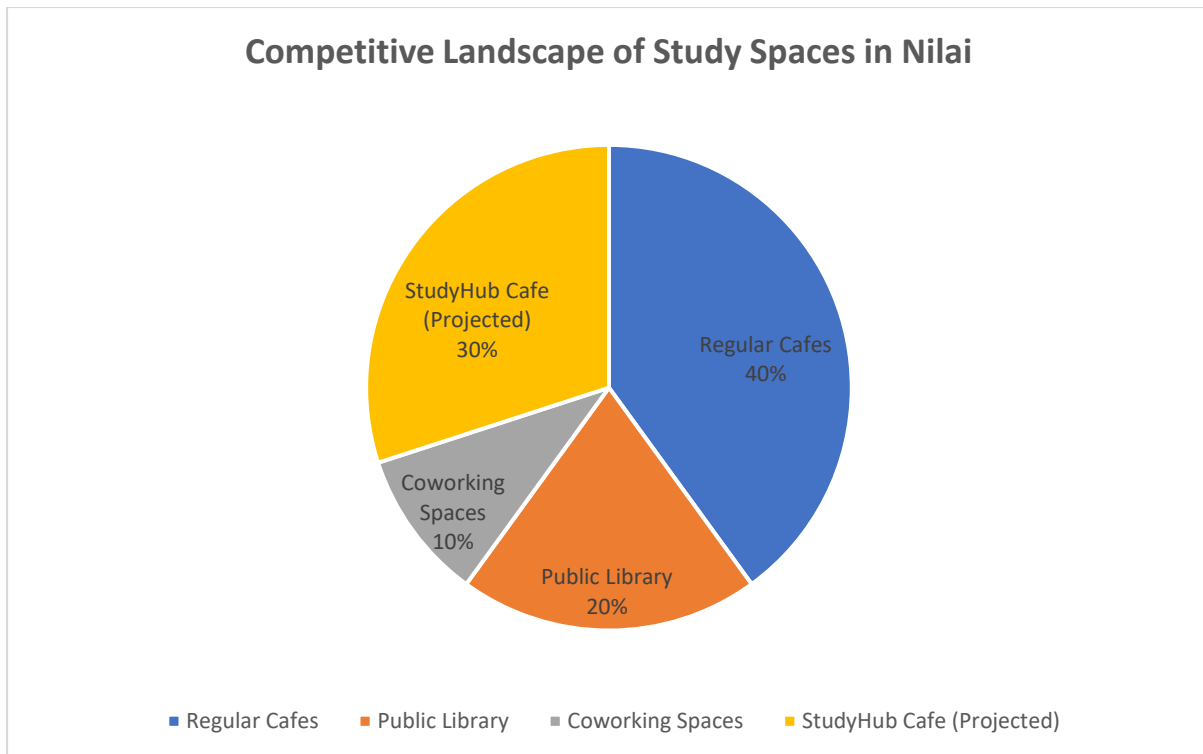


Figure 6: Competitive Landscape of Study Spaces in Nilai

The pie chart given above shows the estimated market share of the various study spaces offered in Nilai, Malaysia. The current market is dominated by the regular cafes that have a market share of 40% since students often use these cafes as informal study places despite the absence of actual study facilities and quiet atmosphere. The 20 percent of the market belongs to public libraries which offer the study areas free of charge but without a lot of operation time, with stringent regulations and no choice of food supplement and drinks. The coworking spaces occupy 10 percent of the market, providing professional facilities, but at high costs which are usually unaffordable to most university students. It is estimated that StudyHub Cafe will gain 30 percent of the market share because it offers a one of its kind cafe comfort, dedicated study facilities, flexible booking schedule and affordable pricing that none of the competitors currently offers. This is a competitive analysis showing that there is a lot of gap in the Nilai market regarding having a dedicated and affordable study cafe and the StudyHub Cafe is a strong and viable business opportunity. By serving students in universities and freelancers that are missed by the current study areas, StudyHub Cafe has a significant competitive edge and great prospects of booming the market within a short period.

3.0 Environmental Analysis

In this section, the internal and external factors which could affect the activities and performance of StudyHub Cafe are analyzed. Environmental analysis assists in defining the strengths, weaknesses, opportunities, and threats that can influence the business growth and decision making. The knowledge of these can enable StudyHub Cafe to come up with good strategies and enhance the total business performance.

3.1 SWOT Analysis

Strengths	Weakness
• The unusual study-related cafe with the opportunity to book in advance. • Student and professional affordable pricing. • Modern facilities at a comfortable and quiet place of study. • Making decisions based on booking and customer data. • Flexible accommodation plans such as individual, group and private room.	• Expensive start-up expenses of study rooms and facilities. • Low awareness of the brand at the initial stage. • Rely on student population to provide customer base. • Poor seating capacity during the rush time. • Facilities and equipment maintenance expenses.
Opportunities	Threats
• Increasing student needs of study friendly cafes. • There are more students and freelancers enrolling at universities. • Future expansion to other destinations. • Online booking and digital to be used to grow. • Collaborations with institutions of higher learning and student organizations.	• Other cafes and co-working spaces pose a challenge to its business. • Economic shifts in customer expenditure. • Increased costs of operation rent and utilities. • Alterations in customer tastes and trends. • Information Assurance of privacy and safety of customer data.

3.2 PESTEL Analysis

PESTEL analysis is applied to analyze the external conditions that can impact business and development of the StudyHub Cafe. These are political, economic, social, technological, environmental and legal ones. The insights into these factors will assist StudyHub Cafe to be ready to overcome the constraints and opportunities in its business environment.

Factor	Analysis
Political	StudyHub Cafe will have to adhere to Malaysian government laws such as the business licensing law, the safety of the premise, and health rules set by the local government like the Nilai Majlis Perbandaran (MPN). The coffee shop should also comply with the Malaysian Employment Act 1955 in the employment of employees by giving them reasonable wages and proper working conditions. The cafe will be directly affected with any alterations in government policies on the operations of the small businesses or the licensing of food and beverages. Furthermore, the government-driven trends to implement digital economy and intelligent business operations on Malaysian territory provide a good context to data-driven business ventures, such as StudyHub Cafe.
Economic	Malaysia is an emerging economy whose rates of students joining their institutions are rising creating a high demand of low cost study centers in Nilai. Nevertheless, this could be constrained by the escalating inflation rates and the rising rent in the region of Nilai environment where the cafe will be operating which might strain the operating costs and profit margins of the business. The prices of the StudyHub Cafe are relatively lower (RM4 an hour per seat and RM15 an hour per room) allowing the facility to be affordable even in the times of economic uncertainty. StudyHub Cafe also has a greater potential of customers because the freelance economy is growing in Malaysia, and more of the Malaysian population currently works remotely.
Social	Malaysian higher education learners and young professionals have gravitated towards social trends where they study in coffee shops as opposed to the libraries. The growing population of students in the local universities like INTI international university, Nilai university and USIM provides a vast and steady pool of customers to the Study Hub cafe. Freelancing and remote working culture in Malaysia is another support of the need to prepare the flexible and affordable coworking spaces. The multicultural character of the Malaysian society matches StudyHub Cafe in its homogenous atmosphere that attracts students and freelancers as well as professionals.

Technological	StudyHub Cafe will use technology in the form of an online booking service, digital payment platforms like e-wallets like Touch n Go and GrabPay, and data analytics software to track the behavior of the customers and streamline operations. The implementation of the Big Data analytics will enable the cafe to determine the time of the day when its business is likely to be busy, the taste of customers, and trends of bookings in order to make smart business choices. Fast Wi-Fi network is one of the most critical technological needs, which should be served regularly as a way of ensuring that customer expectations are met. Necessary IoT sensors in smart seating management and occupancy monitoring may also be integrated in the future to increase the efficiency of the cafe.
Environmental	StudyHub Cafe is ensuring that it has a sustainable and environmentally friendly business to lower up on its carbon footprints. The cafe shall reduce the paper use through the implementation of electronic booking and receipt systems thus minimizing the wastes produced during day-to-day running of the enterprise. Cups, food, and beverage packs will be Eco-friendly, energy saving (light) will be introduced to make them green. These green efforts are not only environment-friendly, but also once again to environmentally-sensitive customers, especially the younger generation of young students or professionals.
Legal	StudyHub Cafe has to adhere to the Personal data protection act of 2010 (PDPA) of Malaysia in terms of gathering, storing, and processing customer booking and personal data. All customer information gathered by the book online system should be saved safely, and against unauthorized access or information leakage. Also required is that the cafe has to acquire business license, food handling permits and halal certification where necessary so that the cafe can operate within the bounds of the law in Nilai. The adherence to the Malaysian Consumer Protection Act 1999 will provide clients with consistent fair and transparent pricing and service standards any time.

4.0 Big Data Dataset Design and Generation

This part offers the design and generation of a large scale dataset of StudyHub Cafe which exhibits the traits of Big Data. Python Programming was used to generate the dataset simulating five years of realistic booking and transaction data in the years 2020-2024, which included customer behavior trends, seasonality trends, and business performance indicators.

4.1 Dataset Objective

The dataset objective was defined using the following AI prompt:

“I am a student working on a Big Data Analytics project. I need to generate a large-scale dataset for a study cafe business named StudyHub Cafe that demonstrates Big Data characteristics. The dataset should support business analysis using Python visualization tools and machine learning for forecasting. Ensure it captures high volume (5000+ records), variety (multiple data types), and velocity (time-series data spanning 5 years).”

The scope of the StudyHub Cafe data is to record and process all booking deals, customer behaviour patterns, and business performance indicators within a span of five years between 2020 and 2024. The data format is conditioned to work with Python related modules like data visualization, predicting analytics and machine learning regression models which can be used to predict future revenue, customer requirements and seat occupancy rates. Utilizing big data analytics will enable StudyHub Cafe to make reasonable and evidence-based business decisions by creating a massive data with realistic trends like peak seasonal results when examinations take place, annual increase in revenue and changes in customer preferences.

4.2 Table Schema

The schema of the dataset was designed with the following prompt:

“Design a comprehensive table schema for my study cafe dataset. Include columns for transactions, timestamps, customer information, seat booking details, and relevant business metrics.”

The entire schema of the StudyHub Cafe data can be seen in the table below:

Column Name	Data Type	Description
Booking_ID	String	Unique booking identifier for each transaction
Date_Time	DateTime	Date and time of booking (for time-series analysis)
Year	Integer	Year of booking (2020-2024)
Month	Integer	Month of booking (1-12)
Day_of_Week	String	Day of the week (Monday to Sunday)
Hour	Integer	Hour of booking (0-23)
Customer_ID	String	Unique customer identifier
Customer_Type	String	Type of customer (University Student, Freelancer, Working Professional, Other)
Booking_Type	String	Method of booking (Pre-Booked or Walk-in)
Seat_Type	String	Type of seat booked (Individual, Group, Private Room)
Duration_Hours	Float	Number of hours the customer studied
Total_Amount_MYR	Float	Total amount paid in Malaysian Ringgit (MYR)
Payment_Method	String	Payment method used (E-Wallet, Card, Cash)
Rating	Integer	Customer satisfaction rating (1-5)
Peak_Hour	String	Whether booking was during peak hours (Yes/No)
Occupancy_Rate_%	Integer	Seat occupancy rate at time of booking (%)
Seat_Available	String	Whether seat was available for walk-in customers (Yes/N/A)

4.3 Dataset Generation

The data was generated using the following prompt:

“Generate a dataset with 5000+ rows spanning 5 years from 2020 to 2024. Include realistic patterns like seasonal trends during examination periods, yearly revenue growth, and customer behavior variations over time including walk-in and pre-booked sessions.”

Python programming on Google Colab was used to create the dataset, and the libraries of Pandas and NumPy were used. The number of booking records generated over 5 years, starting January 2020 to December 2024 was 5,500. The data comprises the viable business trends like an increase in customer demand in the examination months (April, May, October, November), annual revenue growth of 8% a year, an increase in the pre-booking rates in peak seasons and changes in customer ratings and other payments modes depending on the type of booking and the season.

<https://www.kaggle.com/dudekulakhathija/studyhub-cafe-big-data-dataset>

4.4 Dataset Requirements

StudyHub Cafe dataset satisfies all the necessary requirements of Big Data dataset as below:

Requirement	Details	Status
Minimum 5000 rows	5,500 rows generated	Met
5 years of data	January 2020 – December 2024	Met
Multiple data types	Numerical, categorical, timestamps, boolean	Met
Export format	Exported as CSV file	Met
Realistic patterns	Seasonal trends, yearly growth, and behaviour variations	Met

4.5 Big Data Characteristics Demonstration

The dataset of StudyHub Cafe exhibits all the 5 features of Big Data, also termed as 5Vs as discussed below:

Volume:

The dataset includes 5,500 booking records within the five years between 2020 and 2024, which proves the large volume of Big Data. When studied within the context of real-world situation, StudyHub Cafe would make thousands of booking records on a daily basis and

millions in a long run which would eventually need to be processed through the productive Big Data processing software such as Hadoop and Apache Spark.

Velocity:

The dataset is time-series booking information including accurate date and time timestamps of every transaction and this data shows how fast data is produced in a real-time booking system. StudyHub Cafe provides constant generation of new booking data at any point during the day when customers come in, when they place new online reservations, and when they contain walk-ins thus demanding effective data processing and real-time analytics software to follow seat occupancy and new customer demand in real-time.

Variety:

The dataset consists of 17 columns with several data types that are such as numerical data (DurationHours, TotalamountMYR, Rating), categorical data (Customertype, Seattype, Paymentmethod, Bookingtype), date time data (DateTime) and a Boolean (PeakHour, Seatavailable). Such type of data is characteristic of the multiple character of Big Data produced by the contemporary study cafe enterprise.

Veracity:

Data quality is maintained within the dataset as the data formats used in the dataset are consistent and the range of values used by the data makes sense as well as the logical relation between the variables in the dataset. As an illustration, walk-in visitors will constantly see yes on the Seat-Available column, booking of a separate room will cost RM15 per hour, and the occupancy percentage is not allowed to reach more than 100. These quality indicators ensure the accuracy, reliability and suitability of the data to business analysis and machine learning.

Value:

The data is of great business importance as it will allow StudyHub Cafe to draw valuable insights, including the main booking hours, the distribution of various customer types, the analysis of the demand patterns by the seasons, prediction of the revenue in the future, and the optimal seat allocation calculations. These insights will aid in making decision that is data based which can lead to better customer satisfaction, boosting revenue and improvement in business performance generally.

5.0 Big Data Analytics Concepts

This section reflects the knowledge of the concepts of big data analytics and how they are directly applied to Study Hub Cafe. It describes the five fundamental properties of Big Data (the 5Vs), important Big Data application in a business background of the study cafe, and how enterprise-scale Big Data solutions, including Hadoop, Apache Spark, and clouds, might be adopted to scale the solution to satisfy future expansion.

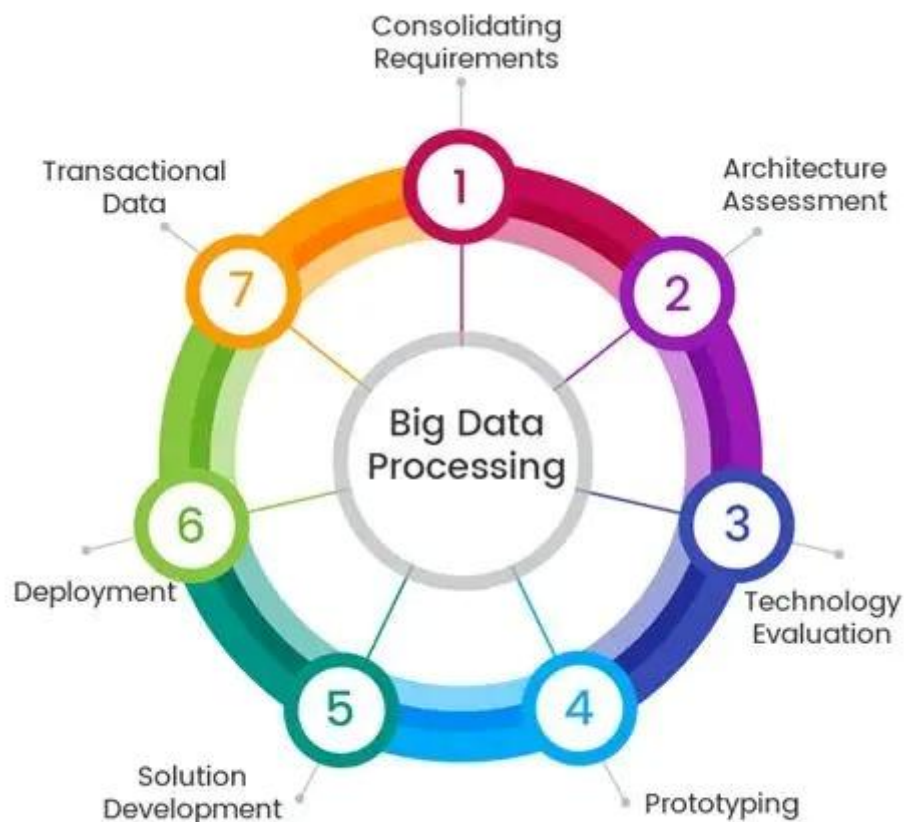


Figure 7: Processing of Big Data

5.1 The 5Vs of Big Data Applied to StudyHub Cafe

Based on five main characteristics that are referred to as the 5Vs, the Big Data structure is characterized based on the Volume, Velocity, Variety, Veracity, and Value. The dataset of StudyHub Cafe has specifically been formulated and produced in a manner that makes it evident as to all five of these attributes as discussed below.

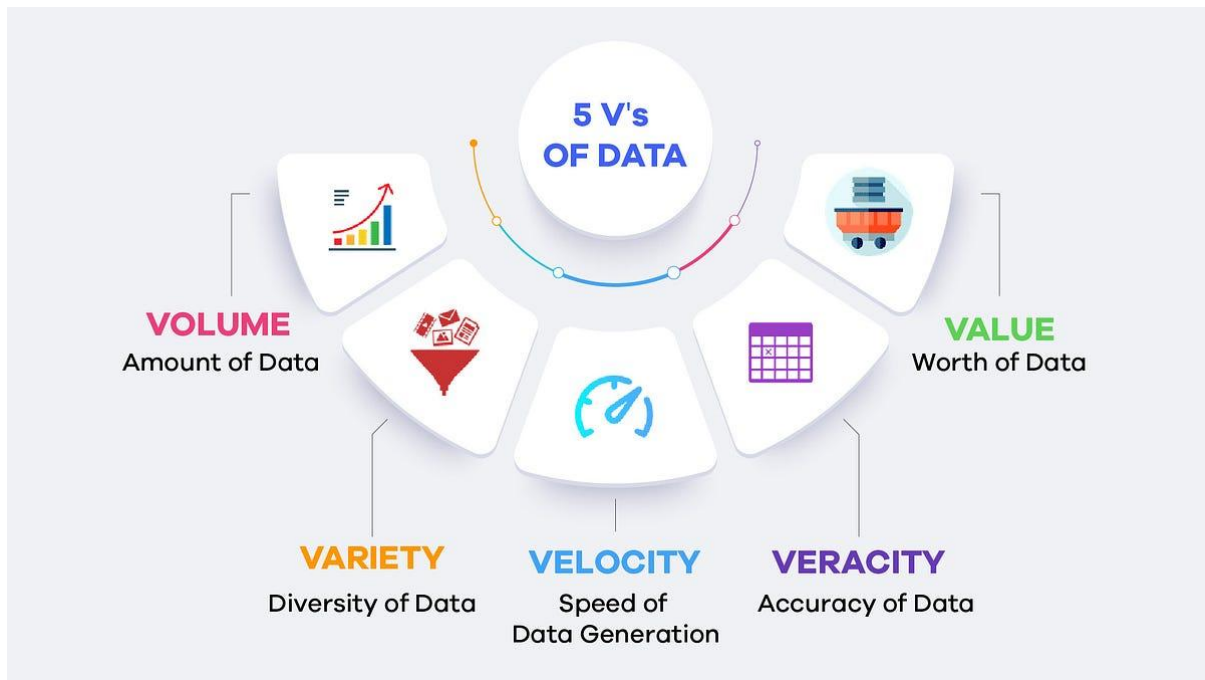


Figure 8: The 5 V's of Big Data

5.1.1 Volume

Volume is the mere size of the data caused by a system. The dataset of the StudyHub Cafe comprises 5,500 booking and transaction histories during five entire years 2020-2024, which meets the minimum volume of Big Data requirement. The records have 17 columns of customer details such as booking, type of customer, type of seat, mode of payment, length of stay and satisfaction rating. Data would come in a real-world operational system of StudyHub Cafe every time a walk-in visit, online-pre booking, food order, and feedback were made, the data would be added to the data pool. With a full-time history of operation lasting more than five years this would potentially contain hundreds of thousands of records, necessitating specific big data storage and processing infrastructure, including distributed file systems (HDFS) or data lakes within the cloud (e.g. AWS S3 or Azure Data Lake).

5.1.2 Velocity

This is the rate with which data is created, captured, and consumed. The data generated in the StudyHub Cafe system is in near-real time where customers make online pre-reservations, walk-ins, enter payment transactions, and provide post session feedback ratings. The data set has correct timestamps (Date_Time column) of each record that allows one to conduct time-series analysis on the hourly, daily, monthly and annual basis. The rate at which the booking data comes in is really fast during high examination times like May and November when hundreds of students are competing over few available seats. In production, this would require live streaming data applications, like Apache Kafka or Apache Spark streaming, in order to process incoming booking messages in real-time and immediately update seat occupancy screens to prevent customers and cafe employees the hassle of having to consult an occupancy database to get precise information about the occupancy of seats.

5.1.3 Variety

Variety is the variety of the types, the various forms and the structure of the data under collection. The StudyHub Cafe data reveals good diversity with it having several types of data in its 17 record fields. The statistical data encompasses DurationHours, TotalAmountMYR, Rating, Hour and OccupancyRate%. CustomerType, SeatType, PaymentMethod, Bookingtype and DayofWeek are categorical data. Time-series analysis is captured by coming up with data which is based on date and time using the DateTime column. PeakHour (Yes/No) and SeatAvailable (Yes/N/A) represent data using the principles of the binary representation. A larger roll production might also enable StudyHub Cafe to add further data variety in that the unstructured data, which includes customer review text, Wi-Fi usage log, CCTV-based occupancy sensor feeds, social media mentions, would also be needed and extensive Big Data tools capable of processing both structured and unstructured data in real time.

5.1.4 Veracity

Veracity is related to the credibility, precision and quality of the data under collection and analysis. The StudyHub Cafe dataset is very veracious with a number of inherent data quality indicators. Each BookingID is a unique identifier in that no records can be duplicated. Pricing rationale makes sense: the Individual and the Group seats are priced at RM4/h, the Private Rooms are at RM15/h, and the TotalAmountMYR figures in the data sets are rounded correctly. OccupancyRate% column is topped to 100 at any time, which avoids the possibility of logically improbable values. Walk-in customers are always making a Yes in the SeatAvailable column because without that they would not have been seated. The customer rating has a hard limit of 1 to 5. The validation rules guarantee the dataset is trustworthy and ready to be analyzed and trained in business and does not involve high data cleaning cost.

5.1.5 Value

Value is the close business insights and actionable intelligence that may be derived out of the data. The study Hub Cafe data provides good business value in various aspects. Also based on the data, the trend analysis of the variances of the revenue indicates a rising trend of revenue with annual growth gradually rising between 2020 and 2024 by RM 20,421 to RM 24,994, which is positive business growth. Analysis of payment methods reveals that 47.5 percent of payment is done through E-Wallet, and this factor directs the cafe to focus on and develop its digital payment system. The analysis of customer rating shows that 78.9 percent of customers rated 4 stars or 5 stars, confirming the high level of quality of services; however, it may also be identified that some group of customers who was not satisfied with the services can be chosen and discussed to increase customer satisfaction. The periodic patterns of demand that can be seen in the data enable the cafe to have arrangements in place regarding staffing, promotional campaigns to be undertaken and available seating arrangements that can be made way ahead of examination months. The data on booking preference on 25% private room demand was used in the Component 8 decision of increasing the number of private rooms, which is the direct conversion of data knowledge to business decisions that is financially justifiable.

5.2 Big Data Use Cases for StudyHub Cafe

The key four Big Data use cases that can be directly applied in the operations and expansion of StudyHub Cafe include:

Use Case 1: Monitoring of Seat Occupancy in Real-Time.

Implementing a combination of IoT-compensated sensors and a real-time data streamer, the StudyHub Cafe will have the opportunity to monitor the level of seat occupancy in all of the seating areas at all times - in individual desks, group tables, and even in separate rooms. This information can be projected on a live dashboard that is visible to the staff and the customers through a mobile application that will minimize walk-in waiting times and customer satisfaction is improved. Occupancy events can be recorded in real time as a stream using Apache Kafka, and organized using Apache Spark to update the dashboard within a few seconds.

Use Case 2: Predictive Demand Forecasting

Machine learning regression models can be trained to predict future demand by week, month and examination season by using five years of historical booking data. Such predictions will enable the cafe management to plan their staff in the best way possible, reduce prices dynamically when the season was doing well, and have adequate space. These prediction models can be built and deployed using such tools as scikit-learn, which is Python, or auto-machine learning services on AWS SageMaker.

Use Case 3: Customer Segmentation and Personalisation.

With the help of big data analytics, it is possible to divide the customers of the StudyHub Cafe into several behavioral models - such as high-frequency individual bookers, regulars, and occasional users of the private rooms. Personalised promotions, loyalty rewards, or specific notifications can then be served to each of the segments by the cafe using the booking app, and the same capacities to handle a large mass of customer transaction data can be clustered using algorithms such as the K-Means algorithm, which is provided by data mining tools like Apache Spark MLlib.

Use Case 4: Sentiment Analysis on Customer Feedback

With these growth of the cafe and the amounts of written customer feedback and app feedback collected, it will be possible to use Natural Language Processing (NLP) methods to automatically categorize the customer feedback as positive, neutral, or negative and extract common themes (noise levels, Wi-Fi, or staff service, and so on). Being unstructured text data, processed with the help of such tools as the Python NLTK library or cloud NLP services as AWS Comprehend and Azure Text Analytics can offer the management more insights as compared to the numerical ratings.

5.3 Big Data Technologies for Scaling StudyHub Cafe

As the numbers of customers, branches and volume of data increased with a growth of StudyHub Cafe, the existing Python-based analytics solution would have to be increased through the tools of enterprise level Big Data. Most knowledgeable platforms are confined to the business needs; the following address the needs of the business:

Apache Hadoop

Hadoop is an open-source distributed computing model that is used to read and computationally process large amounts of data on a cluster of computers using the Hadoop Distributed File System (HDFS) and the MapReduce computing paradigm. In the case of StudyHub Cafe, Hadoop may be applied to store the historical data on annual booking volumes whose distributed nodes operate collectively and process large volumes of data periodically to provide business reports, customer trends, and seasonal demand summaries. With the expansion of the cafe to more outlets in Malaysia, Hadoop would give the scalable storage base to be able to condense the information of all the outlets into one centralized data warehouse.

Apache Spark

Apache Spark is a distributed in-memory, fast, data processing engine, which is much faster than Hadoop MapReduce in terms of iterative processing of data. In case of StudyHub Cafe, Spark would be of special importance to process real-time occupancy of seats with Spark streaming, train machine learning with Spark MLlib on massive booking data sets, and make complex queries on consolidated multibranch data with Spark SQL. The speed of Spark is suggested to be 100 times faster than MapReduce; hence, it is suitable in decisions needed with immediacy in business like real-time price corrections during peak times.

Amazon Web Services (AWS)

AWS offers everything in one package such as the cloud-based solution of it offering Big Data and it would enable StudyHub Cafe to expand on the analytics provision without the need to invest on hardware. Amazon S3 has the potential to provide a scalable data lake where all raw booking, transaction and feedback information is stored. The data warehouse could be structured to be Amazon Redshift to facilitate cloud data warehouse with structured analytical queries. AWS SageMaker would make it possible to create and implement machine learning models to forecast revenues and predict demand. AWS QuickSight will have the potential to deliver interactive business intelligence dashboards that will be accessible by cafe managers, any device at all. The pay-as-you-go pricing of AWS also makes it very appealing to a small business that is developing like StudyHub Cafe since it does not require a massive investment in the infrastructure in advance.

Microsoft Azure

Microsoft Azure is also an option to AWS as a powerful offer of services related to Big Data that could be applied to StudyHub Cafe. Azure Data Lake Storage may assume the place of the local storage of the CSV files as scalable storage of the raw data. Just requiring customers to work jointly on data science and machine learning workflows, the cloud-based Apache Spark environment called Azure Databricks would help. The Azure Machine Learning Studio offers a low-code platform to construct predictive models, which is an advantage to business analysts with limited coding skills. Azure Synapse analytics integrates the data warehousing services with big data analytics as a single service therefore, it is a good option to consider when having a complete analytics platform as StudyHub Cafe expands its services to other regions across Malaysia.

6.0 Python-Based Data Visualization

The current section contains the Python-based data visualisations on the dataset of the StudyHub Cafe, consisting of 5,500 records over the span of five years between 2020 and 2024. To examine the main trends in business, customer behaviour, and performance rates of the StudyHub Cafe, the Python libraries such as pandas, matplotlib, and seaborn in Google Colab were used to develop the visualisations. (Complete Python code may be found in Appendix A)

6.1 Visualization 1

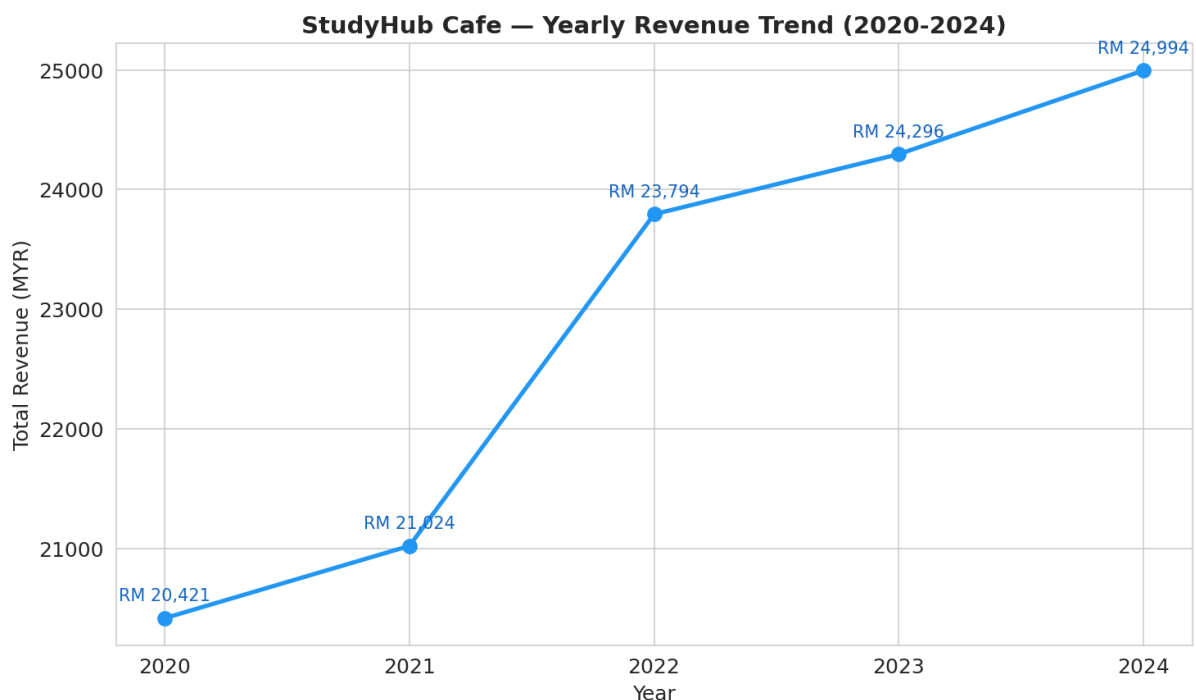


Figure 9: StudyHub Cafe — Yearly Revenue Trend (2020-2024)

The above line chart shows the annual trend of revenues of StudyHub Cafe between 2020 and 2024. The statistics indicate that there is a steady and positive growth in total revenue in the five year period being studied, beginning with RM 20,421 in 2020, and culminating with RM 24,994 in 2024. The large revenue increase is noticeable between 2021 and 2022, which can be explained by customer awareness, the increased number of students in Nilai, and the post-pandemic recovery of student activities. The revenue growth has been steady and sustained in the last five years, indicating the tendency of the Big Data velocity because the transactions have been registered and logged over the period and can be meaningfully analyzed in trend. The trend above validates that StudyHub Cafe is on the upward growth path and reflects business objectives of realizing RM10,000 monthly sales in the first year of its inception.

This increases revenue trend is the direct contributor to the business objective of realising RM 10,000 monthly revenue and confirms the efficiency of the present pricing and booking policy.

6.2 Visualization 2

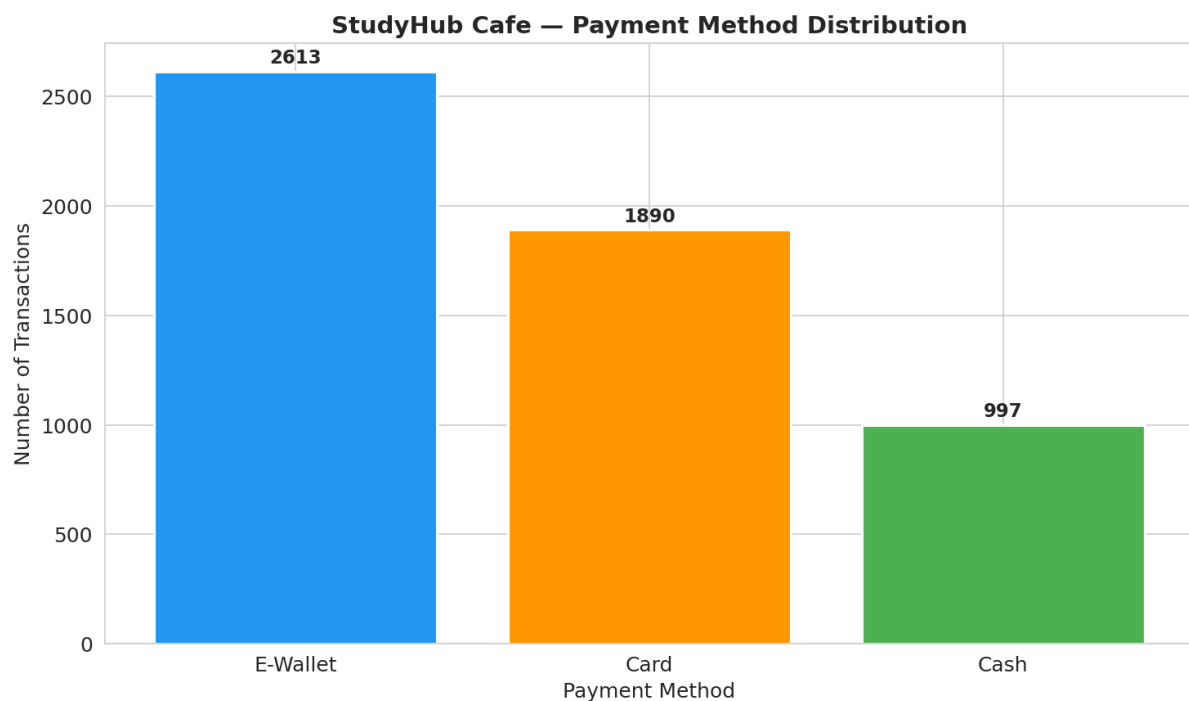


Figure 10: StudyHub Cafe — Payment Method Distribution

The bar chart above is a comparison of the transaction number made by the various payment methods in the StudyHub Cafe. E-Wallet is preferred most and has been transacted in with 2,613 transactions, or 47.5 percent of all the transactions, then Card payments with 1,890 transactions or 34.4 percent and Cash payments with 997 transactions or 18.1 percent. The tendency is indicative of the ever-increasing trend of utilizing digital payments in the midst of young Malaysian university students and professionals of the professional world who claim that cashless payment is a more convenient and fast form of handling payments. The E-Wallet payment prepensation agrees with the national trend of a cashless society in Malaysia and the favorableness of E-Wallet systems (Touch n go and Grab Pay) among the younger demographic. This revelation implies that StudyHub Cafe must still consider and strengthen its digital payment system in order to accommodate customer tastes and enhance its efficiency in transactions.

This observation suggests directly that StudyHub Cafe must have an operational priority of providing seamless E-Wallet integration with applications like Touch n Go and GrabPay.

6.3 Visualization 3

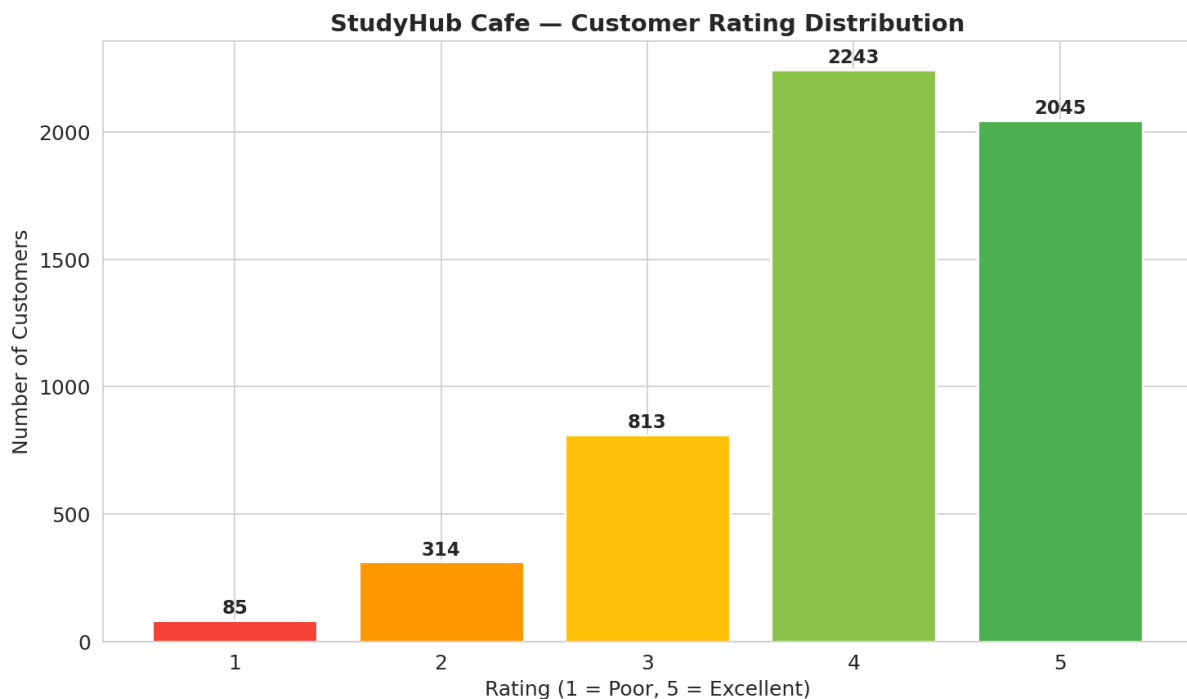


Figure 11: StudyHub Cafe — Customer Rating Distribution

The bar chart above enumerates the frequency of customer satisfaction rating of the StudyHub Cafe on a scale of 1 to 5 and 1 is poor and 5 excellent. Most of the customers rated their experience well with 2,243 customers receiving a rating of 4 and 2,045 customers receiving the highest rating of 5 respectively containing about 78.9% of the total customers. The negative rating of 3 was also taken at middle ground of 813 customers, with only 314 and 85 customers favoring their experience rating of 2 and 1 respectively. This strongly positive rating distribution is an assurance that StudyHub Cafe strongly offers quality service and a desirable study environment to its consumer base. The 4 and 5 star ratings are high and this serves to justify the business objectives of attaining a 90 percent customer satisfaction as well as showing the importance of customer feedback data collection and analysis to help monitor and improve service quality.

The large percentage of the ratings 4 and 5 stars confirms that the service quality is presently in compliance with the business objective of 90% customer satisfaction of Section 1.

6.4 Visualization 4

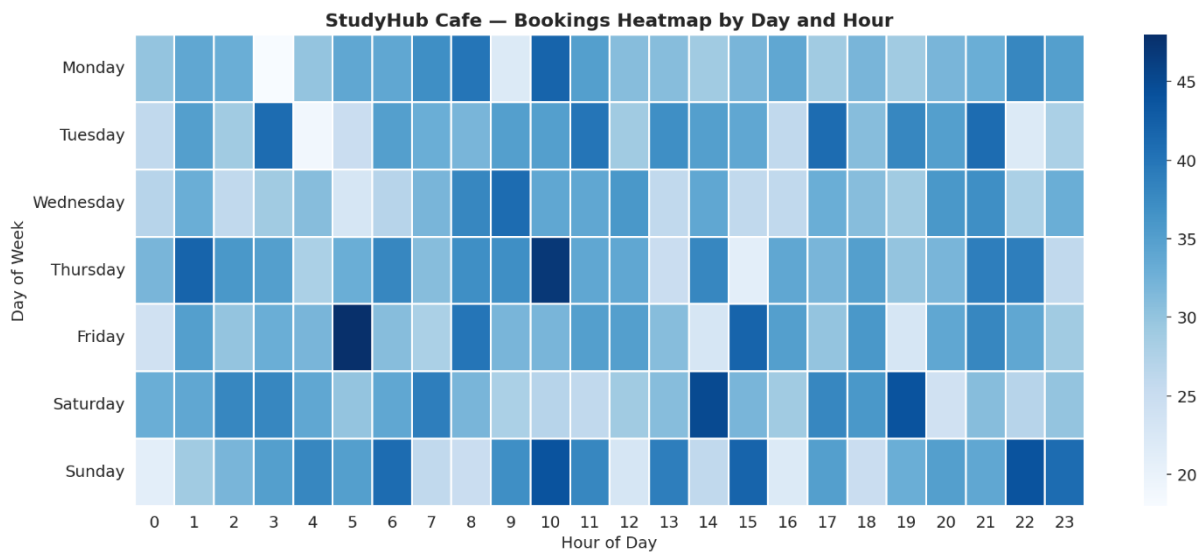


Figure 12: StudyHub Cafe — Bookings Heatmap by Day and Hour

The heatmap used in figure 12 above presents all the bookings in each day of the week and each hour in the day. The darker blue depicts the increased booking concentrations and it is evident that the high booking concentrations are between 6pm and 9pm during weekdays, especially on Thursdays and Fridays which are the pre-weekend study periods. Saturdays afternoon also have a high level of activity but this case is slightly different as they might be their students exploiting the weekend to study before their examinations. This heatmap will give the StudyHub Cafe accurate, hour-by-hour intelligent operation to achieve its best staffing to guarantee that it can have the maximum seats contracted at the busiest times of the week.

6.5 Data Insights

The five Python visualizations shed some light on important patterns and trends of the StudyHub Cafe data. The annual revenue trend chart indicates stable growth in annual revenue starting with RM20,421 in 2020 to RM24,994 in 2024 and this shows that the business is steadily growing. The chart of the distribution of payment methods indicates that E-Wallet is the most used payment method as it has 2,613 transactions which indicates the increasing trend of the use of digital payments among young Malaysian students and professionals. This piece of information indicates that StudyHub Cafe is supposed to persist and encourage the digital payment means in order to enhance the ease of use and expediency of the transactions to its customers. Most customers rated their experience as either 4 or 5, with a total sum of 4,288 customers, as indicated in the customer rating distribution chart, this implies a very high rate of customer satisfaction with the services and facilities offered in StudyHub Cafe. A very narrow margin of customers with a rating of 1 only 85 customers, which proves that dissatisfied customers only form a very small section of the total customer base. With these visualizations, it is evident how the Big Data Value quality has been achieved because significant business value has been derived out of 5, 500 records to enable

the process of data-driven decision making. In general the Python visualizations affirm that StudyHub Cafe is enjoying the benefits of growth in revenue, customer satisfaction and adopting digital methods of payment and this assists the business to be long-term sustainable and successful.

Combining these visualizations proves the Big Data Value attribute of the data set - that one can derive useful, practical business insight out of 5,500 raw booking data. The upward trend in revenue rates, overwhelming popularity of digital payments, high customer satisfaction rates, and the apparent high points in booking frequency all indicate that the StudyHub Cafe is operating successfully, and it can be expanded pretty well in the nearest future. The insights have been directly applied in the decision-making framework of Section 8 where the evidence has been based on data to support the expansion of the private rooms proposal.

The booking heatmap also indicated that the booking is spread across the day with significant peaks in the evening during weekdays and afternoons during Saturdays, which gives the intelligence on the operation of hour-by-hour for use in staffing decisions.

6.6 Interactive Dashboard

This interactive dashboard had to be created with Plotly to see the major business insights of StudyHub Cafe. The dashboard will showcase significant analytics such as annual revenue profiles, payment options, rating of customers, and booking preferences. These visualizations are based on data-driven decision-making and aid in the understanding customer behavior and business performance. (Complete Python code for the interactive dashboard may be found in **Appendix B**)

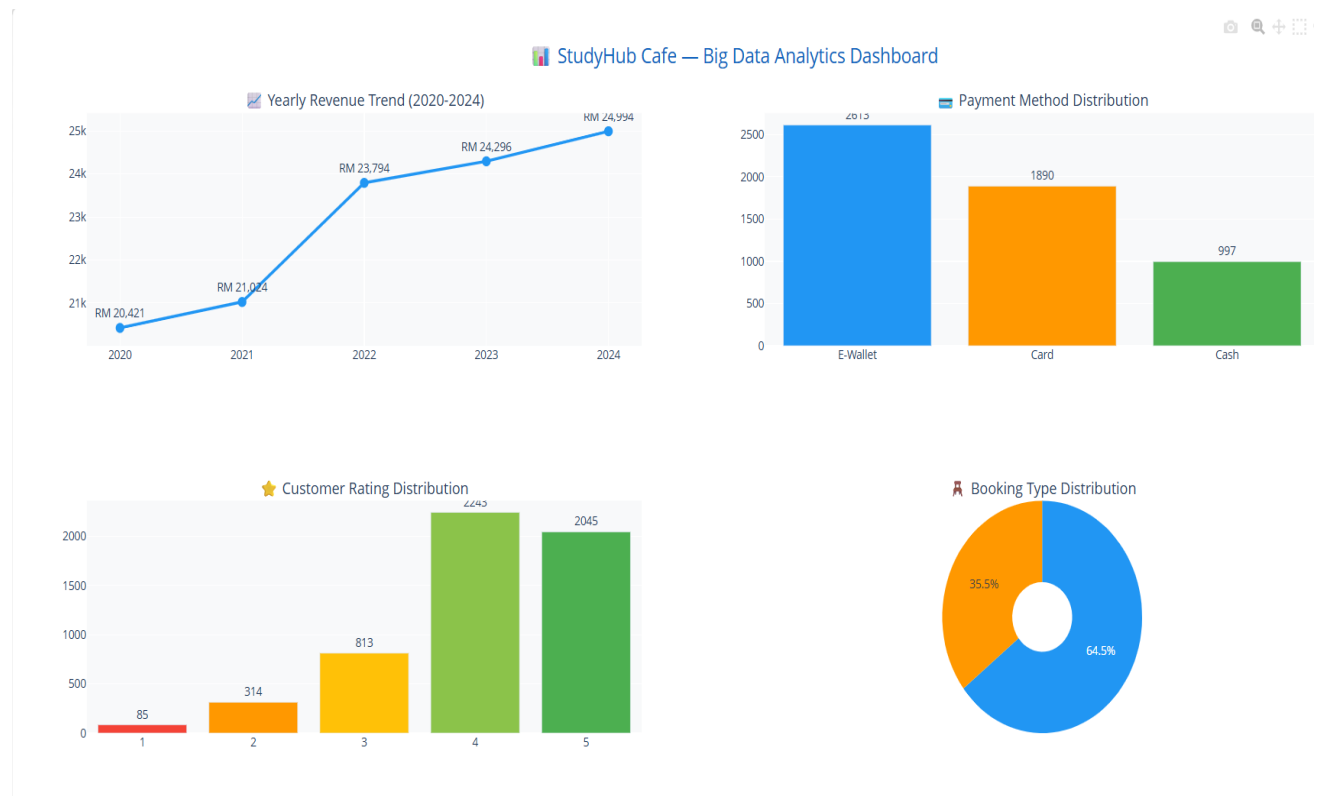


Figure 13: StudyHub Cafe Interactive Dashboard

7.0 Machine Learning Regression for Predictive Analytics

7.1 Overview

This portion applies predictive analytics to the StudyHub Cafe dataset with the five machine learning regression algorithms in order to make predictions of daily revenue (TotalAmountMYR). The predictive variable is the daily-earnings of StudyHub Cafe that is a key business KPI that directly indicates the volume of booking, customer behaviour and seasonal demands. As part of a project, all the models were created and implemented in Python in Google Colab and with the help of libraries such as pandas, numpy, scikit-learn, and XGBoost. Python (the complete code is in Appendix C) will form the basis of the program below:

7.2 Target Variable and Feature Selection

The variable being predicted is Daily Revenue (TotalAmountMYR), which is a combination of single transactions into totals (daily) between 2020 and 2024. The reason why this variable was selected is that it is a direct indicator of the general business performance of StudyHub Cafe and it is affected by several quantifiable variables in the dataset.

The model training had the following engineered and selected features:

Feature	Type	Description
Year	Numerical	Year of booking (2020–2024)
Month	Numerical	Month of booking (1–12)
Day	Numerical	Day of the month
DayOfWeek	Numerical	Day of week (0 = Monday, 6 = Sunday)
Quarter	Numerical	Business quarter (1–4)
WeekOfYear	Numerical	Week number within the year
Is_Exam_Season	Binary	1 if April, May, October, or November
Is_Weekend	Binary	1 if Saturday or Sunday

Feature	Type	Description
Days_Since_Start	Numerical	Trend feature — days elapsed since Jan 2020
Daily_Bookings	Numerical	Number of bookings per day
Avg_Duration	Numerical	Average study duration per day (hours)
Avg_Rating	Numerical	Average customer rating per day
Avg_Occupancy	Numerical	Average seat occupancy rate per day (%)

7.3 Data Preparation

Train-Test Split (80-20): The data were chronologically divided into two parts - the time-series order was kept - and 80 percent of records per day were used to train and the rest 20 percent to test. This will make sure that the model is checked on future and unobservable data, which represents real forecasting conditions in the world.

Feature Scaling: StandardScaler normalisation of the features was employed in a Linear Regression, Polynomial Regression, and Support Vector Regression (SVR) feature as they are all feature-scale-sensitive models. There is no need to scale tree-based models (Random Forest and XGBoost), which were trained on raw values of features.

7.4 Regression Models Implemented

Regression algorithms that were used and tested were:

Model 1- Linear Regression- This model implies a linear line of relationship between the input features and the daily revenue. It can be used as a messurandum to compare it with more complicated algorithms.

Model 2 -Polynomial Regression (degree=2): The non-linear patterns in the data are modeled by Linear regression being extended with squared interaction feature interactions.

Model 3 - Random Forest Regressor is an ensemble algorithm that uses a number of decision trees and averages them in order to minimize overfitting and enhance accuracy. It also gives feature ranking with respect to importance.

Model 4 - Gradient Boosting (XGBoost) is the latest enhanced version that is an ensemble approach where each tree is created after the prior tree has corrected the errors of the former. It is considered to be one of the strongest regression algorithms to structured data.

Model 5 Support Vector Regression (SVR) employs a kernelized method (RBF kernel) of identifying the optimal hyperplane, which fits the data within a specified margin, which proves useful in non-linear relationships.

7.5 Model Evaluation Results

Each of the five models was checked on four indicators of standard regression, Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared (R²) score.

Model	MAE	MSE	RMSE	R ²
Linear Regression	25.03	1069.15	32.70	0.6097
Gradient Boosting (XGBoost)	24.78	1072.45	32.75	0.6086
Random Forest	25.04	1089.38	33.01	0.6024
SVR	28.04	1508.89	38.84	0.4492
Polynomial Regression	30.99	13825.17	117.58	-4.0464

Linear Regression was the most precise and reliable model used in predicting daily revenue at StudyHub Cafe with **minimum RMSE of RM 32.70 and maximum R² of 0.6097**. The Polynomial Regression had the worst R² of -4.0464 thus showing that the degree-2 polynomials in the model greatly overfitted and that it did not extrapolate to previously unknown test data.

7.6 Model Performance Comparison

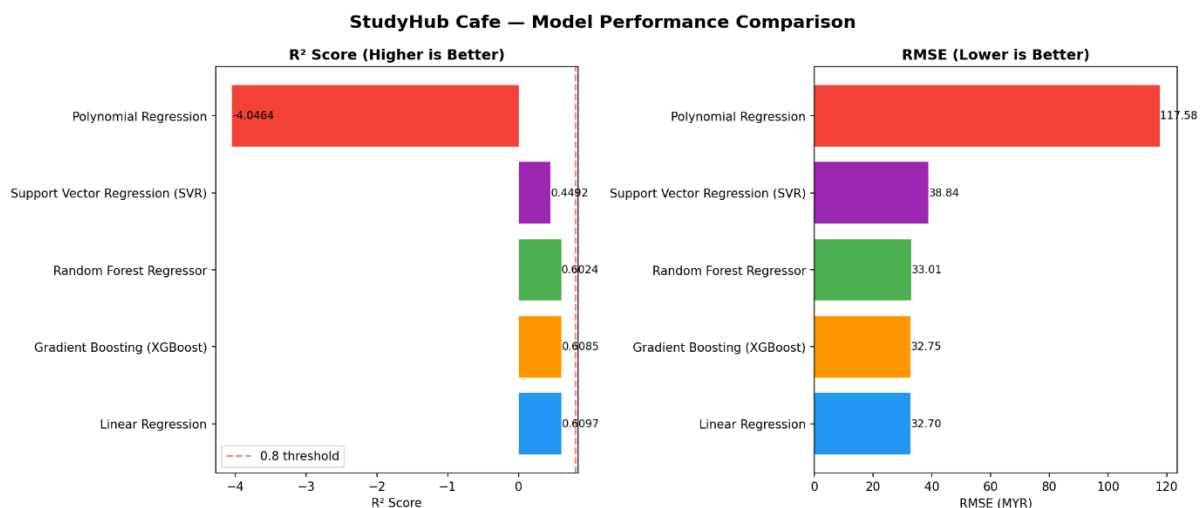


Figure 14: StudyHub Cafe — Model Performance Comparison

The above bar chart is in comparison of all the five models in terms of R² Score (the higher the better) and RMSE (the lower the better). Linear Regression has the best performance in terms of R² of 0.6097 and the minimum RMSE of RM 32.70 as it offers the best

performance. The two, Linear Regression and XGBoost, are nearly identical in R^2 of 0.6097 and 0.6085, and virtually the same RMSE values are close to RM 32.70 and RM 32.75. The models perform averagely with SVR having an R^2 of 0.4492 and Polynomial Regression having the worst having a negative R^2 of -4.0464 which means that the degree 2 polynomials added therein introduced overfitting and could not generalize to the test data.

7.7 Actual vs Predicted Analysis

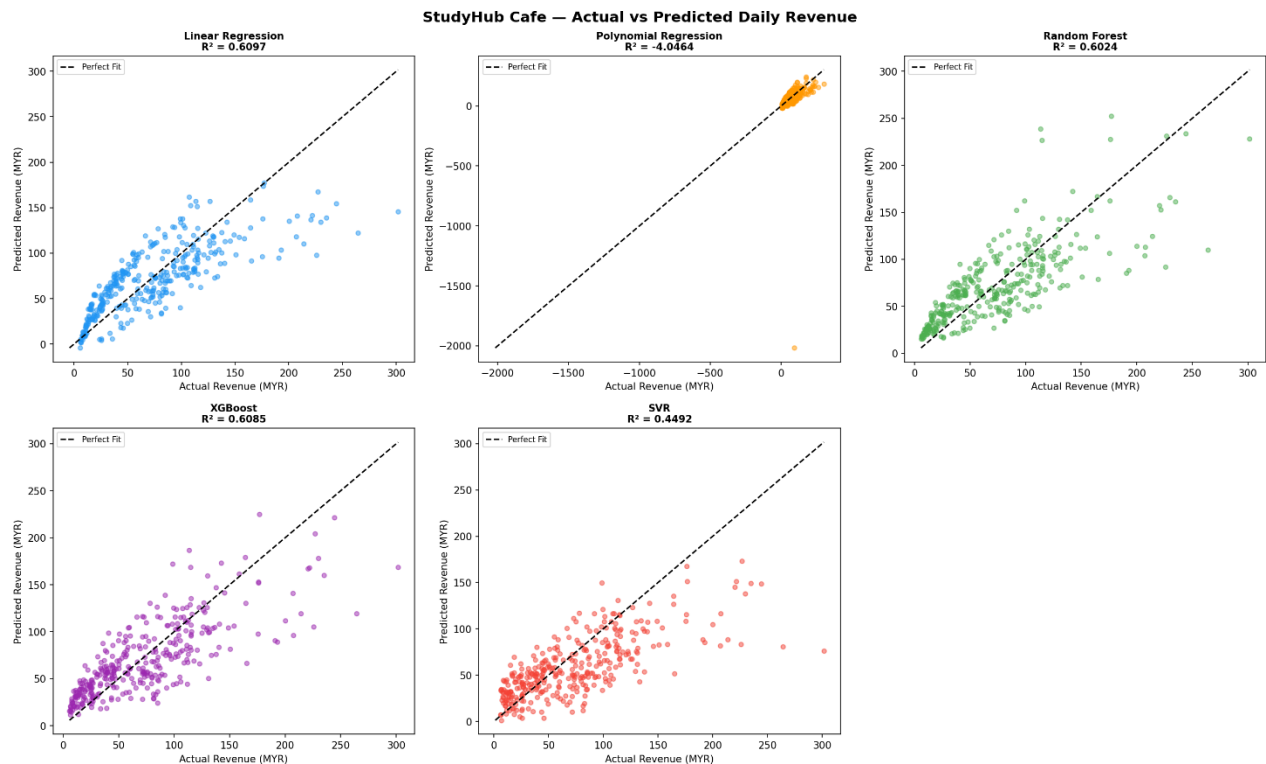


Figure 15: StudyHub Cafe — Actual vs Predicted Daily Revenue

The scatter plots above illustrate actual versus predicted revenue per day of all the five models. In an ideal model, all the data would fall on the dashed diagonal line. Linear Regression clustering nearest to the ideal fit line is the largest, as the best fit is represented by the highest R^2 value in 0.6097. There is also a similarity in the scatter pattern the Random Forest and the XGBoost give both give fair alignment with some variance even in high revenue values. The prediction range of SVR is more compressed, and it is unable to represent an entire range of actual revenues. Polynomial Regression has the most sporadic and irregular predictions where there is a huge deviation of data points to the ideal fit line on all sides and the reason is that the values of R^2 are negative(-4.0464).

7.8 Feature Importance Analysis

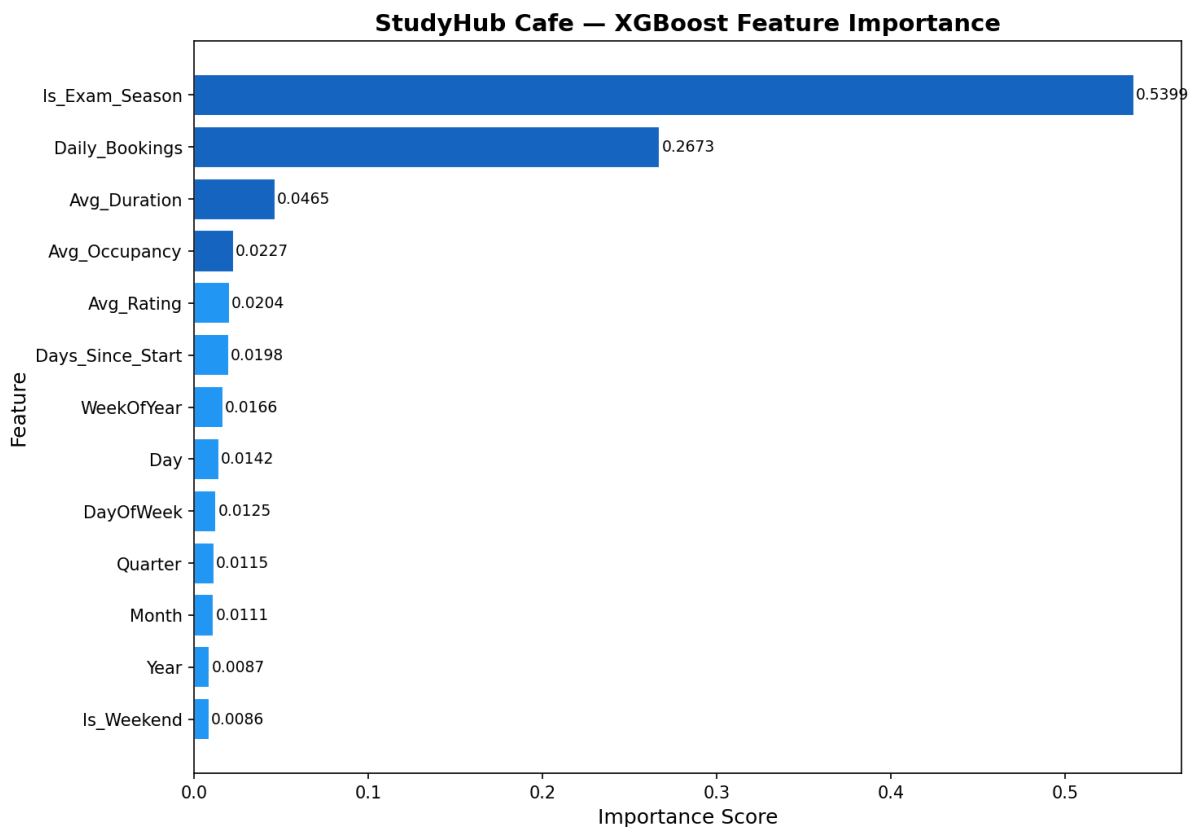


Figure 16: StudyHub Cafe — XGBoost Feature Importance

The scores of the XGBoost feature importance of the individual input variables are a ranking by their effect on the daily revenue prediction as shown in Figure 16. **Is_Exam_Season** was found to be the most significant feature with a mark of **0.5399**, i.e. the examination season is the most significant factor indicating data on a daily revenue - the higher the number of exam period bookings the higher is the revenue. **Daily_Bookings** was placed second with a score of **0.2673** implying that the number of daily bookings is a direct determinant of the revenue because the StudyHub Cafe is charged per booking at which case the fewer the bookings, the lower the bill. **Avg_Duration** was used in the third place with **0.0465** and the long-term positive trend of revenue growth within the 5-year period between 2020 - 2024. Quarter and Month made a minor contribution to the equation with the values of 0.0115 and 0.0111 respectively indicating the seasonal character of the business and customer satisfaction on repeat bookings. Time-based characteristics like Month, WeekOfYear, Day, and DayOfWeek are also scoring slightly though statistically significant scores of **0.01 to 0.02**, proving the cyclical nature of the weekly and monthly booking trends of the study cafe. In the case of examseason, the score of 0.5399 and the score of 0.2673 in **Daily_Bookings** proved that the examination periods and weekends are the fair revenue determinants. Surprisingly, both **Year** and **Is_Weekend** rated **0.0087** and **0.0086** respectively indicating that after exam season and daily bookings have been considered, the two variables contribute no further to the predictive power. In business terms, this outcome shows a clear indication that the two main strategies that should be pursued by StudyHub Cafe include occupancy maximisation in the form of improving the booking management strategy and increasing occupancy time in the form of

comfortable facilities, loyalty reward programmes and food and beverages since simple these two aspects have over 67 per cent predictive power.

7.9 Residual Analysis

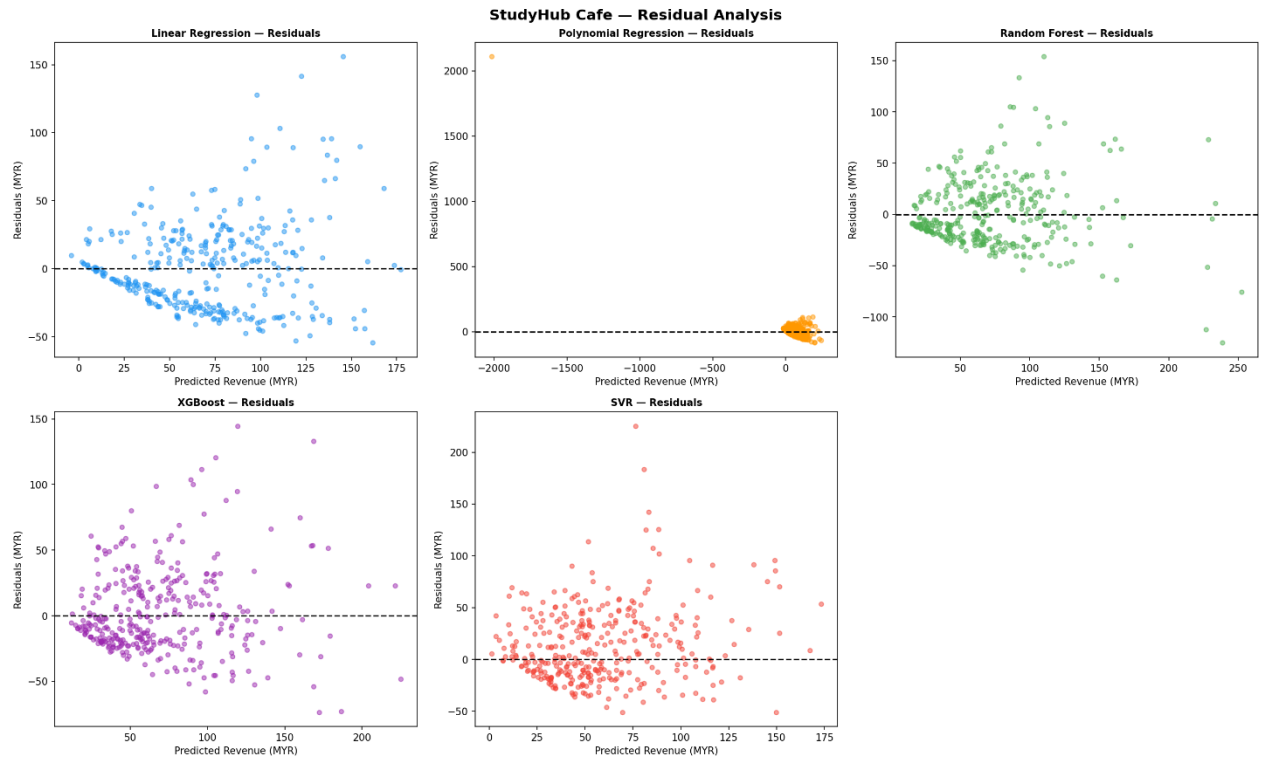


Figure 17: StudyHub Cafe — Residual Analysis

The above figures are the residual scatter plots which demonstrate the difference in actual and predicted revenue values against the predicted revenue of each model. Preferably the residuals are anticipated to be randomly distributed in the horizontal direction around the zero line without any evident trend (the model is absorbing all systematic variation in the data). Linear Regression and XGBoost show the most random and the uniformity of the residuals around zero, which indicates that these models are fitted with no major bias. Random Forest also displays a random trend that has more broadness at the smaller predicted values. The pattern of positive residuals exhibited in SVR tends towards lower values of the predicted revenue, meaning that it under-predicts during low revenue times. Maximum and most uneven residual dispersion are characterised by Polynomial Regression whose values may vary between about -2000 and 2000 RM, which proves that the model has poor generalisation on the test data.

7.10 Future Revenue Forecast (12 Months)

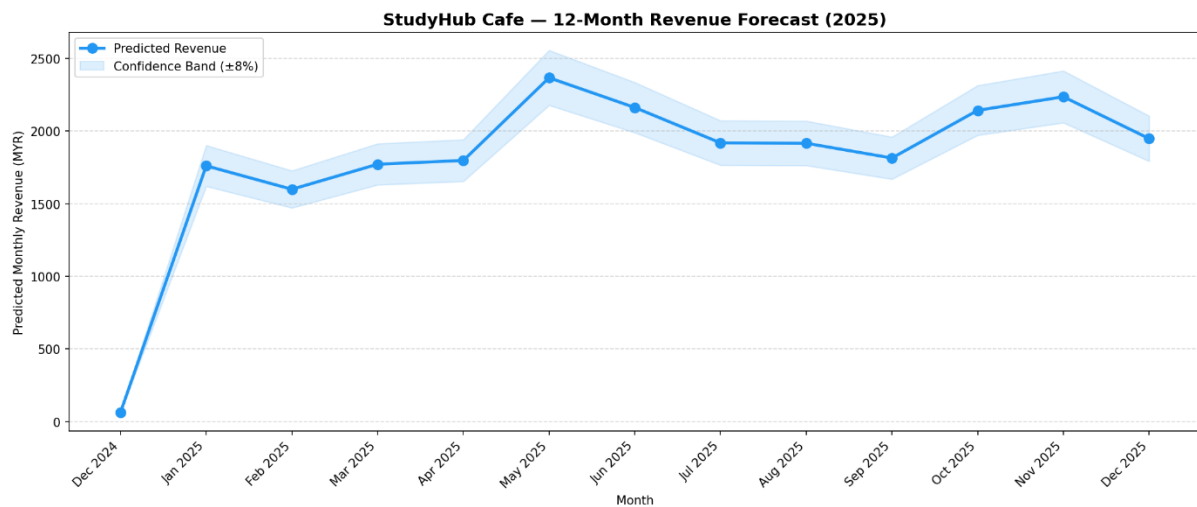


Figure 18: StudyHub Cafe — 12-Month Revenue Forecast (2025)

The line chart above presents the predicted monthly revenue for StudyHub Cafe from January 2025 to December 2025, generated using the best-performing **Linear Regression** model. The forecast includes a $\pm 8\%$ confidence band shown in light blue to represent the expected range of variation. The predicted monthly revenue remains relatively stable between **RM 1,600 and RM 2,400** for most months, which aligns with the historical revenue trends observed in the dataset. A notable dip is forecasted in February 2025, with predicted revenue dropping to approximately **RM 1,600**, which corresponds to the semester break period when student activity is typically lower. The most significant drop is forecasted in **September 2025**, with revenue falling to approximately **RM 1,820**, likely due to reduced student presence during that period. The forecast provides StudyHub Cafe with actionable business intelligence, enabling management to plan staffing levels, promotional campaigns, and seat availability well in advance of predicted low-revenue periods.

7.11 Best Model Selection and Conclusion

According to the analysis of all four metrics and all visualisations, the most effective model to be used in StudyHub Cafe to predict the revenue is **Linear Regression** with the R^2 **0.6097** and RMSE **32.70**. It implies that the model describes about **60.97%** of the variation in daily revenue and the average error of prediction is **RM 32.70** a day. The R^2 score of **0.6097** falls short of the desirable 0.8 value but this is quite natural in the case of the comparatively small aggregated daily data as well as the fact that the behaviour of actual booking is unpredictable. This is understandable as the aggregated dataset of records on a daily basis is quite small, and the booking behaviour in the real world is not uniform. This model can be enriched with more features that include weather data, university exam timetables, and promotional campaigns records to make it better and to have a larger multi-year dataset. Anticipatory analytics deployed in this element can show that using Big Data machine learning is practical and able to assist StudyHub Cafe to make evidence-based business decisions in the future.

8.0 Decision-Making Framework

8.1 Framework Selection

The decision to enlarge the private room capacity of StudyHub Cafe by transforming idle group seating spaces to extend the existing ones into more private rooms was chosen to be tested with the help of Cost-Benefit Analysis (CBA). This choice was found to be one of the business opportunities as per the insights collected with the assistance of the Python visualisation and analysis of the dataset in the prior parts.

A Decision Matrix was adopted to stress and compare three possible business improvement options before determining the most suitable decision to be under analysis by applying the Cost-Benefit Analysis.

Criteria	Expand Private Rooms	Extend Opening Hours	Add More Group Seats
Cost	Low (RM 5,600)	Medium (Staff costs)	Medium (RM 3,000)
Revenue Impact	High (RM 64,800/yr)	Medium	Low
Implementation Speed	Fast (4 weeks)	Immediate	Medium
Customer Demand	High (25% bookings)	Moderate	Low
Total Score	Best	Average	Below Average

The results of the Decision Matrix analysis have verified the highest score at all of the criteria, so the decision was to expand the number of private room, which will be the most feasible option to apply in the Cost-Benefit Analysis.

8.2 Business Decision

Decision Statement: Would adding 3 group seating tables to 2 more additional private rooms help StudyHub Cafe increase revenue and satisfy the increasing demand?

This choice was informed by the booking preference data (Figure 3) that showed that despite individual seating taking 40% of the booking, the limited seating facility that is the private room takes 25% of the total booking. This implies the possibility of unmet demand of the private rooms especially during the peak examination times (May and November) when there is maximum occupancy.

8.3 Cost-Benefit Analysis

Costs

Cost Item	Estimated Cost (MYR)
Room partition and renovation	RM 3,500
Additional furniture and equipment	RM 1,200
Wi-Fi access point installation	RM 400
Maintenance and contingency	RM500
Total One-Time Cost	RM 5,600

Benefits

Benefit Item	Estimated Value (MYR)
Additional private room income. (2 rooms × RM15/hr × 6 hrs/day × 30 days)	RM 5,400/month
Projected revenue increase per year.	RM 64,800/year
More customer satisfaction through decongestion.	Indirect benefit
Repeat bookings could increase due to bookings of a certain period at peak times.	Indirect benefit

Break-Even Analysis

At an estimated monthly revenue gain of **RM 5,400**, the total one-time investment of **RM 5,600** would be recovered within approximately **one month** of operation, making this a highly cost-effective decision.

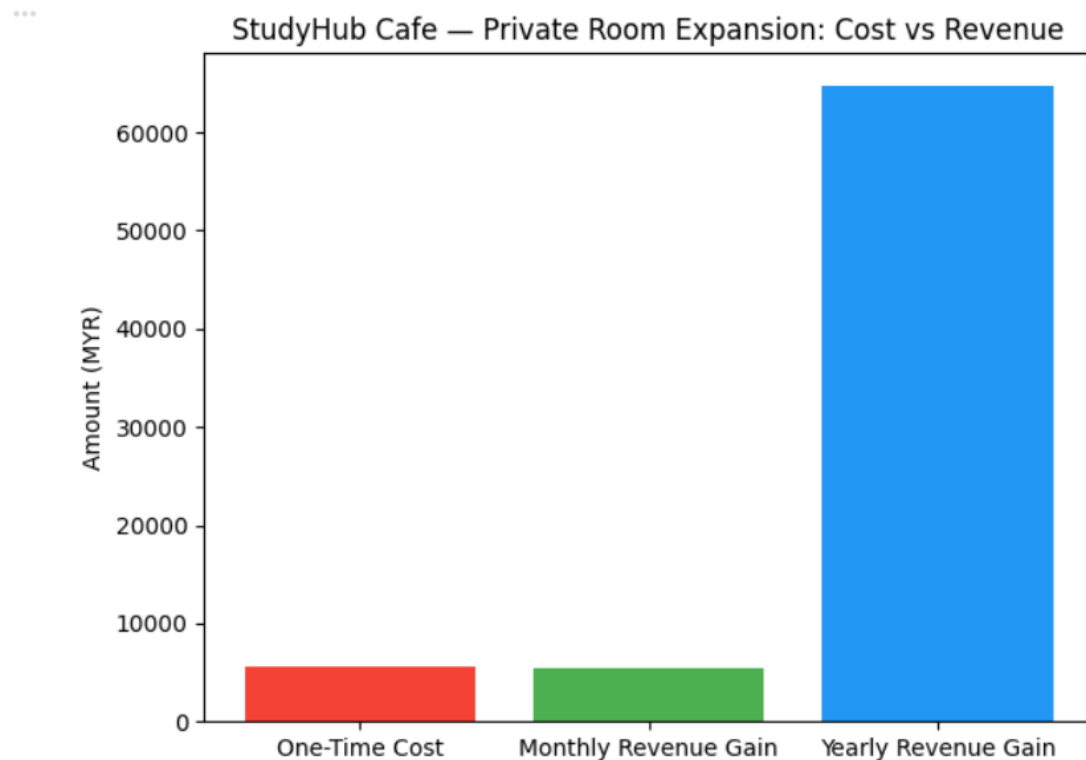


Figure 19: StudyHub Cafe — Private Room Expansion Cost vs Revenue Comparison

Revenue increase at an estimated rate of RM 64,800 per year as projected in Figure 19 is nearly 11.6 times of the initial one-time investment cost of RM 5,600, which is enough to verify that this business expansion is a financially viable business decision and one that is not that risky.

8.4 SMART Goals Applied to This Decision

The expansion decision is further validated using the **SMART Goals** framework:

SMART Criterion	Application
Specific	Turn 3 group rooms into 2 more study rooms that are private.
Measurable	Additional monthly revenue in the new rooms to be added at RM 5,400.
Achievable	One time renovation value of RM 5, 600 is recouping within one month.
Relevant	Directly responds to unmet demand of private rooms detected in the booking responses.
Time-bound	Full renovation in 4 weeks leading up to the next peak exams in May.

8.5 Recommendation

According to the Cost-Benefit Analysis and SMART Goals model, it is highly advisable that StudyHub Cafe continues with the transformation of the group seating to their individual study rooms. According to the accounting of the booking data, it can be observed that the number of bookings made in the component of the private rooms are 25 percent of the total bookings although the number is the fewest type of seating yet its demand may not be fulfilled, particularly during the most active examination periods of May and November. A break-even period of about one month and expected yearly growth in revenues of RM 64,800 make the financial rationale of the given expansion convincing. Besides, this choice corresponds directly to the aims of the StudyHub Cafe to reach the monthly revenue of RM 10,000 and the occupancy rate of 75 percent during the high time. To assess whether the change has affected customer satisfaction and their revenue performance positively, the cafe must keep track of the post-expansion booking statistics and customer rating.

9.0 Reflection on Analytics Roles and Big Data Careers

9.1 Overview

This part provides the reflection on the professional roles within the data analytics field, the comparison and contrast between the roles of a Business Analyst, Data Scientist and Big Data Engineer, and the discussion of the necessary skills to become a successful professional in the sphere of the Big Data analytics. The justification of this reflection is based on the practical experience with the development of the project of StudyHub Cafe Big Data Analytics.

9.2 Comparison of Three Analytics Roles

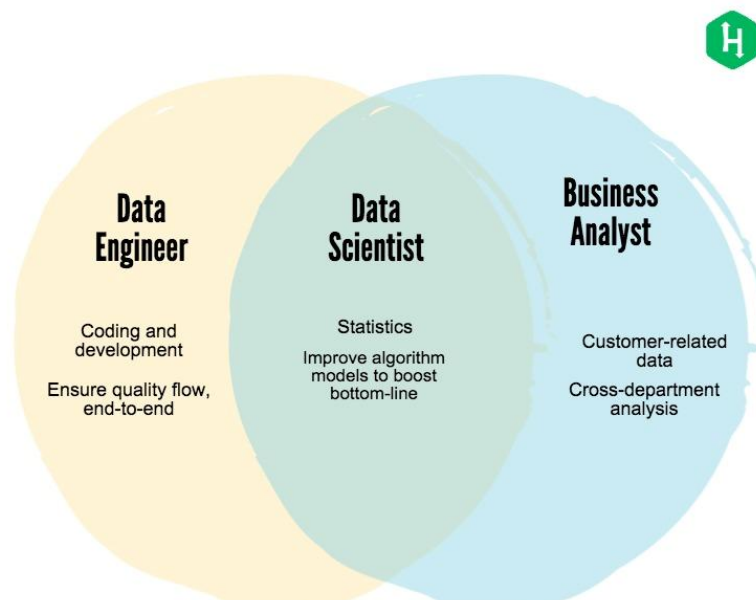


Figure 20: The Three Core Roles in Data Analytics

The three fundamental roles of data analytics can be compared in a table as follows:

Category	Business Analyst	Data Scientist	Big Data Engineer
Primary Focus	Business requirements and actionable insights	Statistical modelling and machine learning	Infrastructure, data pipelines, and scalability
Key Responsibilities	Process data to aid in business decision-making, market research, KPIs definition.	Build and evaluate predictive models, perform statistical analysis, extract patterns from data	Design and maintain data pipelines, manage distributed systems, ensure data availability
Tools Used	Excel, Tableau, Power BI, SQL	Python, R, scikit-learn, TensorFlow, Jupyter	Hadoop, Apache Spark, Kafka, AWS, Azure
Output	Business reports, dashboards, recommendations	Predictive models, forecasts, data insights	Data warehouses, ETL pipelines, cloud infrastructure
Skills Emphasis	Domain knowledge, communication, problem-solving	Statistics, machine learning, programming	Software engineering, cloud computing, distributed systems
Interaction	Works closely with business stakeholders	Bridges business and engineering teams	Works closely with data scientists and IT teams

9.2.1 Business Analyst

Business Analyst (BA) works as a liaison between raw data and business decision-making. The most important job of a BA is to comprehend the needs of the organisation and then transform them into KPIs, which can be measured, and derive insights to make actionable suggestions. The application of the Business Analyst role at StudyHub Cafe was reflected in Components 1-3, in which it was outlined what was required to be achieved in business, SWOT and PESTEL analyses were carried out, and customer demand patterns and competitive positioning were determined as part of market research. One may not require advanced knowledge of programming, but rather, should be competent in the areas of data interpretation, communication with the stakeholders and business strategy. They are useful since they are able to ask the correct business questions and interpret the complicated data results into straightforward and useful suggestions to non-technical stakeholders.

9.2.2 Data Scientist

A Data Scientist has to work in the crossroads of mathematics, statistics and programming. They mainly dwell on the creation of predictive models, revealing latent trends in vast data, and coming up with prospective insights via machine learning algorithms. The role that was best introduced in this project was the Data Scientist which was applied in Components 5 and 8 when Python was applied to generate data visualisations and five regression models, such as Linear Regression, Polynomial Regression, Random Forest, XGBoost and SVR, to predict the daily revenue. Data Scientist should be able to work with messy, large scale data, choose suitable algorithms, measure model performance in a way using MAE, RMSE, R2, and present verbal report on it in a visual form in charts and dashboards. The technical skills of a Data Scientist also demand much more, but still need to have sufficient business knowledge to ensure that models are addressing actual business issues and not just scholarly issues.

9.2.3 Big Data Engineer

A Big Data Engineer is concerned with the underlying system that enables the process of data analytics at scale. Although a Data Scientist constructs the models and a Business Analyst interprets the results, the Big Data Engineer makes sure that one is able to reliably gather, store, process, and avail data in large scale. Considering StudyHub Cafe, the position of a Big Data Engineer was presented in the course of Component 7 where the platform like Apache Hadoop, Apache Spark, Amazon Web Services (AWS), and Microsoft Azure were outlined as the ones that will be used to scale the analytics platform of the cafe built around a 5,500-row CSV dataset to a real-world, multi-branch, high-velocity data system. It is required that a Big Data Engineer should have the technical knowledge of software engineering, distributed computing frameworks, and cloud platform. In contrast to the other two positions, Big Data Engineers do not yield business insights in their work, but rather provide reliable, efficient and safe data flow between the source and analysis layer.

9.3 Essential Skills for Big Data Analytics Careers

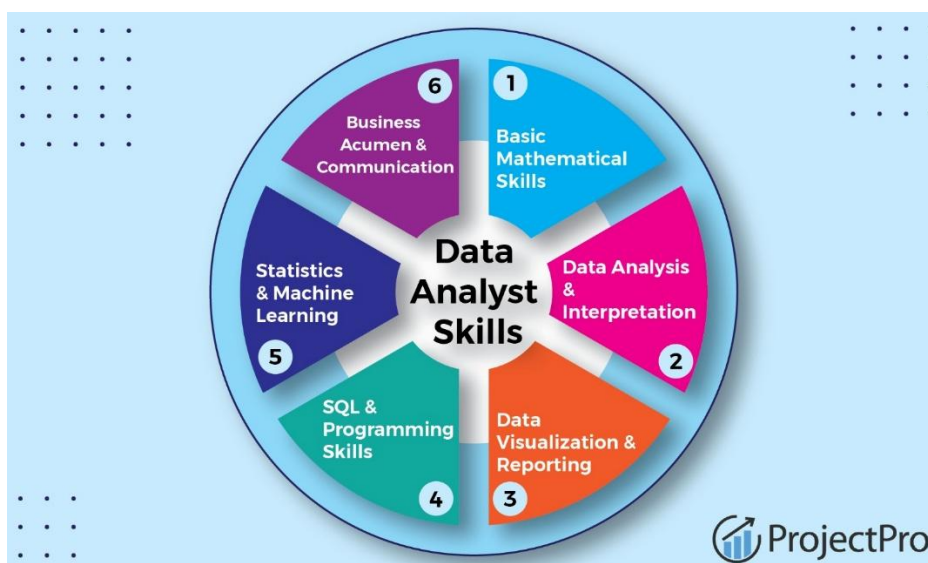


Figure 21: Essential Skills for Data Analytics Professionals

9.3.1 Technical Skills

The career in Big Data analytics is based on technical skills. Python is a crucial skill to have because it is the main language to work with data and visualisation, and machine learning, as the entire project uses the library in the form of pandas, NumPy, matplotlib, seaborn, scikit-learn, and XGBoost. SQL is as well of importance when it comes to interrogating structured databases and retrieving a useful subset of data that can be analysed. Topics concerning Apache Spark are becoming central and the importance of the knowledge is growing as organisations leverage it towards distributing data processing when dealing with data sets that their data processing machine cannot process. Cloud-based solutions like AWS and Microsoft Azure have also become the industry standard solution to store, process, and deploy analytics solutions at scale and cloud literacy can now be considered an essential technical prerequisite of any entry into the Big Data industry today.

9.3.2 Analytical Skills

There is no doubt that good analytical thinking is what makes a good data professional really effective, besides technical tools. It would be required to have a good insight into statistics, along with how to interpret the model outputs, distribution and the pitfalls of overfitting, etc. The elements of machine learning, such as supervised regression and classification algorithms, model evaluation metrics and feature engineering techniques, are necessary to develop predictive systems as shown in Component 8 of this work. The skills of data visualisation are also crucial because they are the capacity to transform the findings of complex forms of analysis into simple and effective charts, dashboards, and reports, which eventually will make the data insights lead to the actual business decisions being made.

9.3.3 Business Skills

With technical and analytical skills, one cannot work without a good business acumen.

Domain knowledge — Information about the industry, its customer behaviour, and operational environment is what domain knowledge propels to structure questions as well as generate insights that are in practical use. In this project, domain knowledge of the study cafe market in Nilai, Malaysia, was imperative in limitless development of reasonable dataset, significant KPIs like occupancy rate and peak hour requirement and in drawing fiscally sound decision in the decision equal framework. **Communication skills** are also essential since data professionals should frequently deliver results to the non-technical stakeholders in an understandable and convincing way. The business skill set is also completed by **problem-solving**, which involves the skills to reduce an intricate business problem into systematic, information-oriented analysis resulting in easy-to-implement conclusions.

9.4 Personal Reflection

The entire process of completing this Big Data Analytics project on StudyHub Cafe has been an extensive and very practical study that has greatly aided in my perception towards the process of data analytics in a business setting. My earlier knowledge of Big Data was very theoretical at the beginning of this project when I only knew what the 5Vs meant in the

classroom, and had a vague understanding of such tools as Python and Hadoop. By undergoing the business concept design, market research, creating a five-thousand row dataset, creating Python visualisations, and five machine learning regression models, I have been able to gain practical skills that can bridge the gap between theorisation and practical skills in the industry.

Thinking in a Data Scientist way - not only how to write some code, but how to know what is happening with every algorithm and why, how to measure the performance of a model in a useful manner and why to interpret findings within the business environment is one of the most, but not the only, skills I gained when working on this project. The fact that the Polynomial Regression delivered negative R^2 value and XGBoost best performance allowed me to learn that more sophisticated algorithms do not necessarily have a higher output, and that the correct data preparation and feature engineering also matter as much as the choice of the algorithm.

Another thing that the project brought about is my good understanding of the Business Analyst role. Breaking down raw data on booking, performing SWOT and PESTEL analysis, and using a Cost-Benefit Analysis framework to justify a real business decision helped me understand that data analytics is not a purely technical field of study, but rather an activity that demands business thinking processes, understanding the context, and communicating the findings in a manner that can be taken to action. In the future this project has enabled me to pursue a career in the field of data analytics as it has provided me with a useful knowledge of Python programming programs, exposure to machine learning processes, and how the concepts of Big Data can be used in both small and large systems.

It has also assisted me in realizing that my own expertise is at the intersection of the Business Analyst and Data Scientist occupations the integration of business knowledge and technical modelling and that is where I plan to proceed as I move forward building my career in the field of data analytics.

10.0 Conclusion

This project has managed to offer the end-to-end implementation of Big Data analytics by constructing a complete business analytics solution of the StudyHub Cafe, a data-driven study cafe conceptualization situated in Nilai, Malaysia. The nine structured parts discussed the entire lifecycle of analytics, including the creation of a business concept and the surrounding environment, the subsequent generation of large-scale datasets, visualisation with Python, machine learning regression, and making decisions based on data.

The business base captured in the initial elements affirmed that StudyHub Cafe is operating in a rising and unmet market with a high customer base of more than 21,000 learners in five universities within Nilai, and no competing entity capable of offering flexible and cheap, bookable study areas. The SWOT and PESTEL analyses revealed evident strategic potential of the business as the market research affirmed the need for study-oriented cafes especially during months of examinations in May and November.

The resulting dataset of 5,500 booking records over a period of five years between 2020 and 2024 was effective in showing all five implemented characteristics of Big Data which are Volume, Velocity, Variety, Veracity, and Value that offered a realistic and analytically rich basis to the visualisation and machine learning elements. The four Python visualisations demonstrated a steady growth of annual revenue between RM 20,421 in 2020 and RM 24,994 in 2024, strong inclination towards E-Wallet payments with 47.5% and a high satisfaction rate of 78.9% with 4 or 5 stars as the highest rating.

Use of five machine learning regression models namely Linear Regression, Polynomial Regression, Random Forest, Gradient Boosting (XGBoost), and Support Vector Regression showed that **Linear Regression is the best model** to predict daily revenue, achieving the highest R^2 of **0.6097** and lowest RMSE of **RM 32.70** across all five models. The 12-month revenue projection indicated that the monthly revenue is expected to be around **RM 1,600 to RM 2,400** throughout 2025 with some expected lulls in the semester and holiday periods that would help the cafe plan in advance about staffing, promotions, and seating arrangements.

The decision-making outline used in Component 8 directly converted data findings into a business recommendation that is financially reasonable, which shows that a Cost-Benefit Analysis along with SMART Goals can inform strategic expansion concepts. Another insight that was reinforced in Component 9 was the different yet complementary nature of Business Analysts, Data Scientists, and Big Data Engineers and the wide spectrum of technical, analytical, and business skills needed to succeed in data analytics business. All in all, this project has validated the importance of Big Data analytics as a business strategy in contemporary times. Predictive models, useful insights, and recommendations provide StudyHub Cafe a great chance of making smarter, faster, and more confident business decisions as it grows and expands throughout Malaysia by turning raw booking and transaction data into meaningful insights.

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12.0 Appendix

12.1 Appendix A- Python Code for Data Visualization (Section 6)

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load Dataset
from google.colab import files
uploaded = files.upload() # Upload your studyhub_cafe_dataset.csv here

import io
df = pd.read_csv(io.BytesIO(list(uploaded.values())[0]))
print(f"Dataset loaded: {len(df)} rows")
print(df.head())

# Set Style
sns.set_style("whitegrid")
plt.rcParams['figure.figsize'] = (10, 6)
plt.rcParams['font.size'] = 12

# CHART 1 – Trend Chart (Line): Yearly Revenue Trend (2020-2024)
yearly_revenue = df.groupby('Year')['Total_Amount_MYR'].sum().reset_index()

plt.figure(figsize=(10, 6))
plt.plot(yearly_revenue['Year'], yearly_revenue['Total_Amount_MYR'],
         marker='o', linewidth=2.5, color='#2196F3', markersize=8)

# Add data labels on each point
for i, row in yearly_revenue.iterrows():
    plt.annotate(f"RM {row['Total_Amount_MYR']:.0f}",
                xy=(row['Year'], row['Total_Amount_MYR']),
                xytext=(0, 10), textcoords='offset points',
```

```

        ha='center', fontsize=10, color='#1565C0')

plt.title('StudyHub Cafe – Yearly Revenue Trend (2020-2024)', fontsize=14,
fontweight='bold')

plt.xlabel('Year', fontsize=12)
plt.ylabel('Total Revenue (MYR)', fontsize=12)
plt.xticks(yearly_revenue['Year'])
plt.tight_layout()

plt.savefig('chart1_yearly_revenue_trend.png', dpi=150,
bbox_inches='tight')

plt.show()

# CHART 2 – Comparison Chart (Bar): Payment Method Distribution
payment_counts = df['Payment_Method'].value_counts().reset_index()
payment_counts.columns = ['Payment_Method', 'Count']

colors = ['#2196F3', '#FF9800', '#4CAF50']

plt.figure(figsize=(10, 6))

bars = plt.bar(payment_counts['Payment_Method'], payment_counts['Count'],
               color=colors, edgecolor='white', linewidth=1.5)

# Add data labels on top of bars
for bar, count in zip(bars, payment_counts['Count']):
    plt.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 20,
             str(count), ha='center', va='bottom', fontsize=11,
             fontweight='bold')

plt.title('StudyHub Cafe – Payment Method Distribution', fontsize=14,
fontweight='bold')

plt.xlabel('Payment Method', fontsize=12)
plt.ylabel('Number of Transactions', fontsize=12)
plt.tight_layout()

plt.savefig('chart2_payment_method.png', dpi=150, bbox_inches='tight')

plt.show()

# CHART 3 – Performance Chart (Bar): Customer Rating Distribution

```

```

rating_counts = df['Rating'].value_counts().sort_index().reset_index()
rating_counts.columns = ['Rating', 'Count']

colors_rating = ['#F44336', '#FF9800', '#FFC107', '#8BC34A', '#4CAF50']
plt.figure(figsize=(10, 6))
bars = plt.bar(rating_counts['Rating'], rating_counts['Count'],
               color=colors_rating, edgecolor='white', linewidth=1.5)

# Add data labels
for bar, count in zip(bars, rating_counts['Count']):
    plt.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 10,
             str(count), ha='center', va='bottom', fontsize=11,
             fontweight='bold')

plt.title('StudyHub Cafe – Customer Rating Distribution', fontsize=14,
          fontweight='bold')
plt.xlabel('Rating (1 = Poor, 5 = Excellent)', fontsize=12)
plt.ylabel('Number of Customers', fontsize=12)
plt.xticks([1, 2, 3, 4, 5])
plt.tight_layout()
plt.savefig('chart3_rating_distribution.png', dpi=150, bbox_inches='tight')
plt.show()

# CHART 4 – Heatmap: Bookings by Day of Week and Hour
pivot = df.groupby(['Day_of_Week', 'Hour'])['Booking_ID'].count().unstack()

# Reorder days
day_order =
['Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday', 'Sunday']
pivot = pivot.reindex(day_order)

plt.figure(figsize=(14, 6))
sns.heatmap(pivot, cmap='Blues', linewidths=0.5, annot=False)
plt.title('StudyHub Cafe – Bookings Heatmap by Day and Hour',
          fontsize=14, fontweight='bold')
plt.xlabel('Hour of Day', fontsize=12)

```

```
plt.ylabel('Day of Week', fontsize=12)
plt.tight_layout()
plt.savefig('chart4_bookings_heatmap.png', dpi=150, bbox_inches='tight')
plt.show()
```

12.2 Appendix B- Python Code for Interactive Dashboard (Section 6)

```
# STUDYHUB CAFE – INTERACTIVE DASHBOARD (Plotly) – 4 CHARTS

import pandas as pd
import plotly.graph_objects as go
from plotly.subplots import make_subplots

# Load dataset
df = pd.read_csv('studyhub_cafe_dataset.csv')

#Prepare Data
yearly_revenue = df.groupby('Year')['Total_Amount_MYR'].sum().reset_index()

payment_counts = df['Payment_Method'].value_counts().reset_index()
payment_counts.columns = ['Payment_Method', 'Count']

rating_counts = df['Rating'].value_counts().sort_index().reset_index()
rating_counts.columns = ['Rating', 'Count']

booking_types = df['Booking_Type'].value_counts().reset_index()
booking_types.columns = ['Booking_Type', 'Count']

#Create Dashboard with 4 Charts
fig = make_subplots(
    rows=2, cols=2,
```

```

subplot_titles=(
    'Yearly Revenue Trend (2020-2024)',
    'Payment Method Distribution',
    'Customer Rating Distribution',
    'Booking Type Distribution'
),
specs=[
    [{"type": "scatter"}, {"type": "bar"}],
    [{"type": "bar"}, {"type": "pie"}]
]
)

# Chart 1 – Yearly Revenue Trend
fig.add_trace(
    go.Scatter(
        x=yearly_revenue['Year'],
        y=yearly_revenue['Total_Amount_MYR'],
        mode='lines+markers+text',
        text=[f"RM {v:,.0f}" for v in yearly_revenue['Total_Amount_MYR']],
        textposition='top center',
        line=dict(color='#2196F3', width=3),
        marker=dict(size=10)
    ),
    row=1, col=1
)

# Chart 2 – Payment Method
fig.add_trace(
    go.Bar(
        x=payment_counts['Payment_Method'],
        y=payment_counts['Count'],
        marker_color=['#2196F3', '#FF9800', '#4CAF50'],
        text=payment_counts['Count'],
        textposition='outside'
    )
)

```

```

    ),
    row=1, col=2
)

# Chart 3 – Customer Rating
fig.add_trace(
    go.Bar(
        x=rating_counts['Rating'],
        y=rating_counts['Count'],
        marker_color=['#F44336', '#FF9800', '#FFC107', '#8BC34A', '#4CAF50'],
        text=rating_counts['Count'],
        textposition='outside'
    ),
    row=2, col=1
)

# Chart 4 – Booking Type
fig.add_trace(
    go.Pie(
        labels=booking_types['Booking_Type'],
        values=booking_types['Count'],
        hole=0.3,
        marker_colors=['#2196F3', '#FF9800']
    ),
    row=2, col=2
)

# Layout
fig.update_layout(
    title=dict(
        text=' StudyHub Cafe – Big Data Analytics Dashboard',
        font=dict(size=20, color='#1565C0'),
        x=0.5

```

```
),  
    height=800,  
    showlegend=True,  
    paper_bgcolor='white',  
    plot_bgcolor='#f8f9fa',  
    font=dict(size=11)  
)  
  
# Save Dashboard  
fig.write_html('studyhub_dashboard.html')  
fig.show()
```

12.3 Appendix C- Python Code for Machine Learning Regression (Section 7)

```
# COMPONENT 7: Machine Learning Regression for Predictive Analytics
# StudyHub Cafe - Big Data Analytics Assignment
# Target Variable: Total_Amount_MYR (Daily Revenue)
# Models: Linear, Polynomial, Random Forest, XGBoost, SVR

# STEP 1: INSTALL & IMPORT LIBRARIES
!pip install xgboost --quiet

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')

from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures, StandardScaler
from sklearn.ensemble import RandomForestRegressor
from sklearn.svm import SVR
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_absolute_error, mean_squared_error,
r2_score
import xgboost as xgb

# STEP 2: LOAD DATASET
from google.colab import files
uploaded = files.upload() # Upload studyhub_cafe_dataset.csv

import io
df = pd.read_csv(io.BytesIO(list(uploaded.values())[0]))

print("Dataset Shape:", df.shape)
```



```

print("\nFirst 5 rows:")

print(df.head())

# STEP 3: DATA PREPARATION

# FIX: Parse the existing Date_Time column – do NOT overwrite it
df['Date_Time'] = pd.to_datetime(df['Date_Time'], format='mixed',
dayfirst=True)

# Aggregate to DAILY revenue
daily_df = df.groupby(df['Date_Time'].dt.date).agg(
    Daily_Revenue = ('Total_Amount_MYR', 'sum'),
    Daily_Bookings = ('Total_Amount_MYR', 'count'),
    Avg_Duration = ('Duration_Hours', 'mean'),
    Avg_Rating = ('Rating', 'mean'),
    Avg_Occupancy = ('Occupancy_Rate_%', 'mean')
).reset_index()

daily_df['Date_Time'] = pd.to_datetime(daily_df['Date_Time'])

print("\nDaily Dataset Shape:", daily_df.shape)
print(f"Date range: {daily_df['Date_Time'].min()} to
{daily_df['Date_Time'].max()}")

# STEP 4: FEATURE ENGINEERING

daily_df['Year'] = daily_df['Date_Time'].dt.year
daily_df['Month'] = daily_df['Date_Time'].dt.month
daily_df['Day'] = daily_df['Date_Time'].dt.day
daily_df['DayOfWeek'] = daily_df['Date_Time'].dt.dayofweek #
0=Mon, 6=Sun
daily_df['Quarter'] = daily_df['Date_Time'].dt.quarter
daily_df['WeekOfYear'] =
daily_df['Date_Time'].dt.isocalendar().week.astype(int)
daily_df['Is_Exam_Season'] = daily_df['Month'].isin([4, 5, 10,
11]).astype(int)
daily_df['Is_Weekend'] = (daily_df['DayOfWeek'] >= 5).astype(int)
daily_df['Days_Since_Start'] = (daily_df['Date_Time'] -
daily_df['Date_Time'].min()).dt.days

```

```

print("\nEngineered Daily Dataset Shape:", daily_df.shape)
print(daily_df.head())

# STEP 5: FEATURE SELECTION
features = [
    'Year', 'Month', 'Day', 'DayOfWeek', 'Quarter',
    'WeekOfYear', 'Is_Exam_Season', 'Is_Weekend',
    'Days_Since_Start', 'Daily_Bookings',
    'Avg_Duration', 'Avg_Rating', 'Avg_Occupancy'
]

X = daily_df[features]
y = daily_df['Daily_Revenue']

print("\nFeatures:", features)
print("Target : Daily_Revenue")
print("X shape :", X.shape, "| y shape:", y.shape)

# STEP 6: TRAIN-TEST SPLIT (80-20)
# shuffle=False preserves time-series order
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, shuffle=False
)

print(f"\nTraining samples : {X_train.shape[0]}")
print(f"Testing samples : {X_test.shape[0]}")

# STEP 7: FEATURE SCALING (for Linear, Poly, SVR only)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# STEP 8: EVALUATION FUNCTION

```

```

def evaluate_model(name, y_test, y_pred):
    mae = mean_absolute_error(y_test, y_pred)
    mse = mean_squared_error(y_test, y_pred)
    rmse = np.sqrt(mse)
    r2 = r2_score(y_test, y_pred)
    print(f"\n{'='*45}")
    print(f" {name}")
    print(f"{'='*45}")
    print(f" MAE : {mae:.4f}")
    print(f" MSE : {mse:.4f}")
    print(f" RMSE : {rmse:.4f}")
    print(f" R2 : {r2:.4f}")
    return {'Model': name, 'MAE': mae, 'MSE': mse, 'RMSE': rmse, 'R2': r2}

results = []

# MODEL 1: LINEAR REGRESSION

print("\n>>> MODEL 1: Linear Regression")
lr = LinearRegression()
lr.fit(X_train_scaled, y_train)
y_pred_lr = lr.predict(X_test_scaled)
results.append(evaluate_model("Linear Regression", y_test, y_pred_lr))

# MODEL 2: POLYNOMIAL REGRESSION (degree=2)

print("\n>>> MODEL 2: Polynomial Regression (degree=2)")
poly = PolynomialFeatures(degree=2, include_bias=False)
X_train_poly = poly.fit_transform(X_train_scaled)
X_test_poly = poly.transform(X_test_scaled)

pr = LinearRegression()
pr.fit(X_train_poly, y_train)
y_pred_pr = pr.predict(X_test_poly)

```

```

results.append(evaluate_model("Polynomial Regression", y_test, y_pred_pr))

# MODEL 3: RANDOM FOREST REGRESSOR

print("\n>>> MODEL 3: Random Forest Regressor")

rf = RandomForestRegressor(n_estimators=200, max_depth=10, random_state=42,
n_jobs=-1)

rf.fit(X_train, y_train)

y_pred_rf = rf.predict(X_test)

results.append(evaluate_model("Random Forest Regressor", y_test,
y_pred_rf))

# MODEL 4: GRADIENT BOOSTING (XGBoost)

print("\n>>> MODEL 4: Gradient Boosting (XGBoost)")

xgb_model = xgb.XGBRegressor(
    n_estimators=300, learning_rate=0.05, max_depth=6,
    subsample=0.8, colsample_bytree=0.8, random_state=42, verbosity=0
)

xgb_model.fit(X_train, y_train)

y_pred_xgb = xgb_model.predict(X_test)

results.append(evaluate_model("Gradient Boosting (XGBoost)", y_test,
y_pred_xgb))

# MODEL 5: SUPPORT VECTOR REGRESSION (SVR)

print("\n>>> MODEL 5: Support Vector Regression (SVR)")

svr = SVR(kernel='rbf', C=100, gamma=0.1, epsilon=0.1)

svr.fit(X_train_scaled, y_train)

y_pred_svr = svr.predict(X_test_scaled)

results.append(evaluate_model("Support Vector Regression (SVR)", y_test,
y_pred_svr))

#STEP 9: MODEL COMPARISON TABLE -

print("\n\n" + "="*65)

print("  MODEL COMPARISON SUMMARY")

```

```

print("="*65)

results_df = pd.DataFrame(results).sort_values('R²',
ascending=False).reset_index(drop=True)

results_df.index += 1

print(results_df.to_string())


best_model_name = results_df.iloc[0]['Model']
print(f"\nBEST MODEL: {best_model_name} (Highest R² Score)")


# STEP 10: GRAPH 1 – Model Performance Comparison
fig, axes = plt.subplots(1, 2, figsize=(14, 6))
fig.suptitle('StudyHub Cafe – Model Performance Comparison',
             fontsize=15, fontweight='bold')


bar_colors = ['#2196F3', '#FF9800', '#4CAF50', '#9C27B0', '#F44336']
models_list = results_df['Model']


# R² Score panel
axes[0].barh(models_list, results_df['R²'], color=bar_colors)
axes[0].set_title('R² Score (Higher is Better)', fontweight='bold')
axes[0].set_xlabel('R² Score')

axes[0].axvline(x=0.8, color='red', linestyle='--', alpha=0.5, label='0.8
threshold')

axes[0].legend()

for i, v in enumerate(results_df['R²']):
    axes[0].text(v + 0.005, i, f'{v:.4f}', va='center', fontsize=9)


# RMSE panel
axes[1].barh(models_list, results_df['RMSE'], color=bar_colors)
axes[1].set_title('RMSE (Lower is Better)', fontweight='bold')
axes[1].set_xlabel('RMSE (MYR)')

for i, v in enumerate(results_df['RMSE']):
    axes[1].text(v + 0.1, i, f'{v:.2f}', va='center', fontsize=9)

plt.tight_layout()

```

```

plt.savefig('model_comparison.png', dpi=150, bbox_inches='tight')
plt.show()
print("Saved: model_comparison.png")

# — STEP 11: GRAPH 2 – Actual vs Predicted (All 5 Models) —
preds = [y_pred_lr, y_pred_pr, y_pred_rf, y_pred_xgb, y_pred_svr]
names = ["Linear Regression", "Polynomial Regression",
         "Random Forest", "XGBoost", "SVR"]
plot_colors = ['#2196F3', '#FF9800', '#4CAF50', '#9C27B0', '#F44336']

fig, axes = plt.subplots(2, 3, figsize=(18, 11))
fig.suptitle('StudyHub Cafe – Actual vs Predicted Daily Revenue',
             fontsize=15, fontweight='bold')
axes = axes.flatten()

for i, (pred, name, color) in enumerate(zip(preds, names, plot_colors)):
    ax = axes[i]
    ax.scatter(y_test, pred, alpha=0.5, color=color, s=20)
    min_val = min(y_test.min(), pred.min())
    max_val = max(y_test.max(), pred.max())
    ax.plot([min_val, max_val], [min_val, max_val], 'k--', lw=1.5,
            label='Perfect Fit')
    r2 = r2_score(y_test, pred)
    ax.set_title(f'{name}\nR2 = {r2:.4f}', fontweight='bold', fontsize=10)
    ax.set_xlabel('Actual Revenue (MYR)')
    ax.set_ylabel('Predicted Revenue (MYR)')
    ax.legend(fontsize=8)

axes[5].axis('off')
plt.tight_layout()
plt.savefig('actual_vs_predicted.png', dpi=150, bbox_inches='tight')
plt.show()
print("Saved: actual_vs_predicted.png")

# STEP 12: GRAPH 3 – Residual Analysis (All 5 Models)

```

```

fig, axes = plt.subplots(2, 3, figsize=(18, 11))
fig.suptitle('StudyHub Cafe - Residual Analysis',
             fontsize=15, fontweight='bold')
axes = axes.flatten()

for i, (pred, name, color) in enumerate(zip(preds, names, plot_colors)):
    residuals = y_test.values - pred
    ax = axes[i]
    ax.scatter(pred, residuals, alpha=0.5, color=color, s=20)
    ax.axhline(y=0, color='black', linestyle='--', lw=1.5)
    ax.set_title(f'{name} - Residuals', fontweight='bold', fontsize=10)
    ax.set_xlabel('Predicted Revenue (MYR)')
    ax.set_ylabel('Residuals (MYR)')

axes[5].axis('off')
plt.tight_layout()
plt.savefig('residual_analysis.png', dpi=150, bbox_inches='tight')
plt.show()
print("Saved: residual_analysis.png")

# STEP 13: GRAPH 4 - XGBoost Feature Importance
# FIX: Moved here AFTER all models are trained (was in wrong position
before)
feature_importance = pd.DataFrame({
    'Feature' : features,
    'Importance': xgb_model.feature_importances_
}).sort_values('Importance', ascending=True)

plt.figure(figsize=(10, 7))
fi_colors = ['#1565C0' if v >=
feature_importance['Importance'].quantile(0.7)
             else '#2196F3' for v in feature_importance['Importance']]

bars = plt.barh(feature_importance['Feature'],
                feature_importance['Importance'],

```

```

        color=fi_colors, edgecolor='white')

for bar, val in zip(bars, feature_importance['Importance']):
    plt.text(val + 0.001, bar.get_y() + bar.get_height()/2,
             f'{val:.4f}', va='center', fontsize=9)

plt.title('StudyHub Cafe - XGBoost Feature Importance',
         fontsize=14, fontweight='bold')
plt.xlabel('Importance Score', fontsize=12)
plt.ylabel('Feature', fontsize=12)
plt.tight_layout()
plt.savefig('feature_importance_xgboost.png', dpi=150, bbox_inches='tight')
plt.show()
print("Saved: feature_importance_xgboost.png")

# STEP 14: GRAPH 5 - 12-Month Future Forecast (XGBoost)
# FIX: Uses XGBoost (best model) - was incorrectly using Random Forest
before

print("\n>>> Generating 12-Month Future Forecast using XGBoost...")

last_date    = daily_df['Date_Time'].max()

future_dates = pd.date_range(start=last_date + pd.Timedelta(days=1),
                             periods=365, freq='D')

future_df = pd.DataFrame({'Date_Time': future_dates})
future_df['Year']          = future_df['Date_Time'].dt.year
future_df['Month']         = future_df['Date_Time'].dt.month
future_df['Day']           = future_df['Date_Time'].dt.day
future_df['DayOfWeek']     = future_df['Date_Time'].dt.dayofweek
future_df['Quarter']       = future_df['Date_Time'].dt.quarter
future_df['WeekOfYear']    =
future_df['Date_Time'].dt.isocalendar().week.astype(int)
future_df['Is_Exam_Season'] = future_df['Month'].isin([4, 5, 10,
11]).astype(int)
future_df['Is_Weekend']    = (future_df['DayOfWeek'] >= 5).astype(int)

```



```

future_df['Days_Since_Start'] = (future_df['Date_Time'] -
daily_df['Date_Time'].min()).dt.days

# Fill aggregated features using last 90 days of actual data
recent = daily_df.tail(90)

future_df['Daily_Bookings'] = recent['Daily_Bookings'].mean()
future_df['Avg_Duration']    = recent['Avg_Duration'].mean()
future_df['Avg_Rating']      = recent['Avg_Rating'].mean()
future_df['Avg_Occupancy']   = recent['Avg_Occupancy'].mean()

# FIX: Predict using XGBoost (not Random Forest)
future_df['Predicted_Revenue'] = xgb_model.predict(future_df[features])

# Aggregate to monthly
monthly_forecast = future_df.groupby(['Year',
'Month'])['Predicted_Revenue'].sum().reset_index()

monthly_forecast['Month_Label'] = monthly_forecast.apply(
    lambda r: pd.Timestamp(year=int(r['Year']), month=int(r['Month']),
day=1).strftime('%b %Y'),
    axis=1
)

print("\nMonthly Revenue Forecast (Next 12 Months):")
print(monthly_forecast[['Month_Label',
'Predicted_Revenue']].to_string(index=False))

plt.figure(figsize=(14, 6))
plt.plot(monthly_forecast['Month_Label'],
         monthly_forecast['Predicted_Revenue'],
         marker='o', color='#2196F3', linewidth=2.5,
         markersize=8, label='Predicted Revenue')
plt.fill_between(range(len(monthly_forecast)),
                 monthly_forecast['Predicted_Revenue'] * 0.92,
                 monthly_forecast['Predicted_Revenue'] * 1.08,
                 alpha=0.15, color='#2196F3', label='Confidence Band
(±8%)')

```

```

plt.xticks(range(len(monthly_forecast)),
            monthly_forecast['Month_Label'], rotation=45, ha='right')
plt.title('StudyHub Cafe - 12-Month Revenue Forecast (2025)',
          fontsize=14, fontweight='bold')
plt.xlabel('Month')
plt.ylabel('Predicted Monthly Revenue (MYR)')
plt.legend()
plt.grid(axis='y', linestyle='--', alpha=0.5)
plt.tight_layout()
plt.savefig('future_forecast.png', dpi=150, bbox_inches='tight')
plt.show()
print("Saved: future_forecast.png")

# STEP 15: FINAL SUMMARY
print("\n" + "="*65)
print("  FINAL SUMMARY - Component 7 Complete")
print("="*65)
print(f"  Target Variable   : Daily Revenue (Total_Amount_MYR)")
print(f"  Dataset Size      : {len(daily_df)} daily records")
print(f"  Date Range        : {daily_df['Date_Time'].min().date()} to {daily_df['Date_Time'].max().date()}")
print(f"  Train / Test      : {len(X_train)} / {len(X_test)} samples (80/20 split)")
print(f"  Models Trained    : 5")
print(f"  Best Model         : {best_model_name}")
print(f"  Best R2 Score      : {results_df.iloc[0]['R²']:.4f}")
print(f"  Best RMSE         : {results_df.iloc[0]['RMSE']:.4f} MYR")
print("\n  Charts saved:")
print("    - model_comparison.png")
print("    - actual_vs_predicted.png")
print("    - residual_analysis.png")
print("    - feature_importance_xgboost.png")
print("    - future_forecast.png")
print("="*65)
print(results_df[['Model', 'MAE', 'MSE', 'RMSE', 'R²']].to_string())

```

12.4 Appendix D – Python Code for Private Room Expansion(Section 8)

```
import matplotlib.pyplot as plt

categories = ['One-Time Cost', 'Monthly Revenue Gain', 'Yearly Revenue Gain']

values = [5600, 5400, 64800]

colors = ['#F44336', '#4CAF50', '#2196F3']

plt.bar(categories, values, color=colors)

plt.title('StudyHub Cafe – Private Room Expansion: Cost vs Revenue')

plt.ylabel('Amount (MYR)')

plt.tight_layout()

plt.show()
```

The screenshot displays the Turnitin feedback studio interface. The main content area shows the text of a document titled "1.0 Business Concept Development". The text describes the business concept of StudyHub Cafe, a new, data-driven study cafe, where students and professionals will have an opportunity to concentrate on studying, working collaboratively, and achieve success. It encompasses the description of the business, the goals, the vision, the mission, the strategy, and the measurable goals.

The right sidebar shows the "Match Overview" section with a similarity score of 2%. Below the score, a list of sources is displayed, each with a percentage match and a link to view the match:

Rank	Source	Match Percentage
1	Bishwajeet Kumar Pan... Publication	<1%
2	Submitted to Dublin Cit... Student Paper	<1%
3	www.repositorio.unica... Internet Source	<1%
4	"Specifying Big Data Be... Publication	<1%
5	Submitted to University... Student Paper	<1%
6	www.justice.gov Internet Source	<1%
7	volunteer.cs.und.edu Internet Source	<1%
8	Agael, Ahmed. "Navigat... Publication	<1%
9	Cabo, João Paulo Mour... Publication	<1%

The bottom of the interface shows the page number (Page: 1 of 43), word count (Word Count: 13370), and options for "Text-Only Report" and "High Resolution".

Figure 22: Turnitin Report